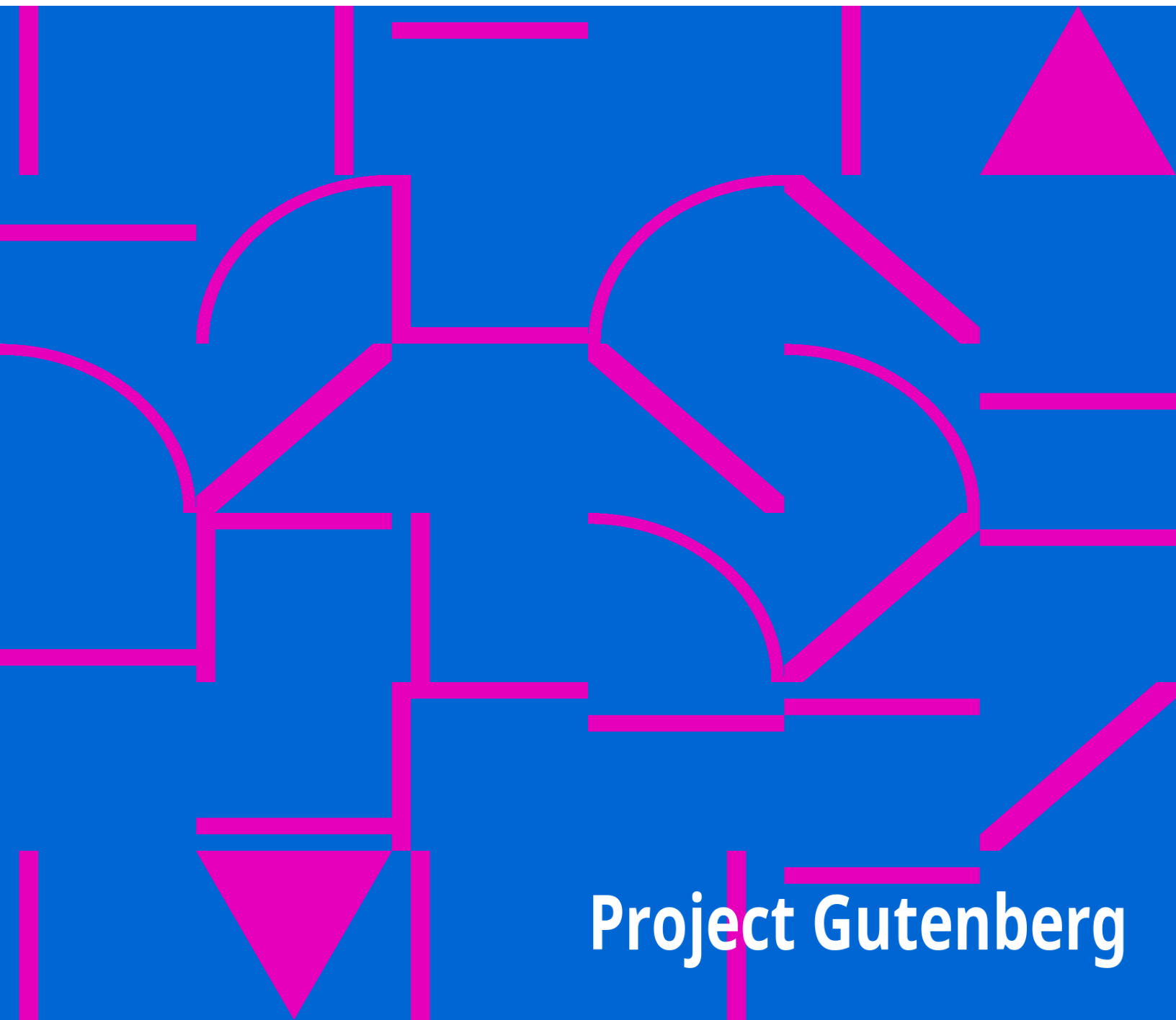


The Baculum in Microtine Rodents

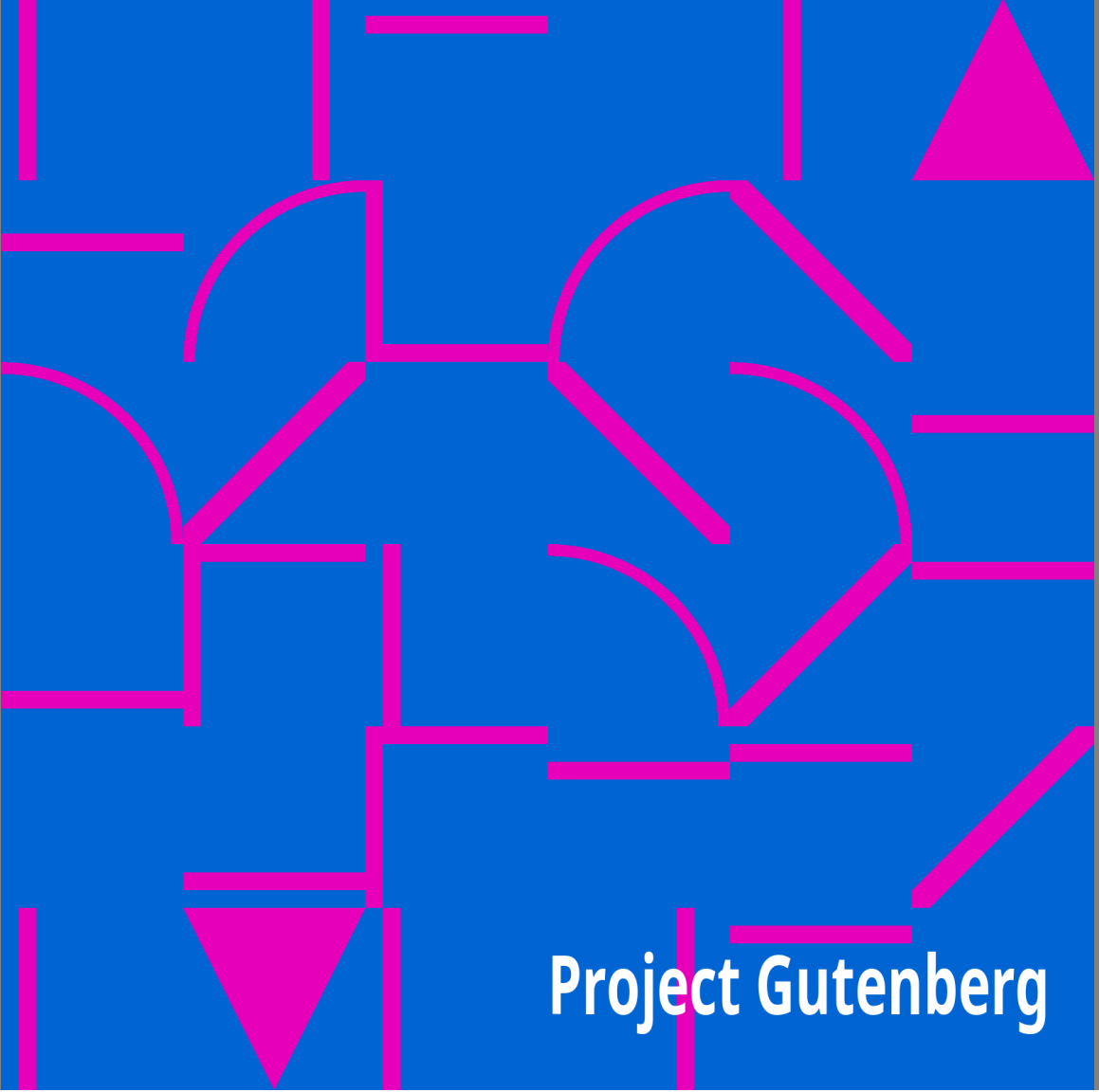
Sydney Anderson

The background of the lower half of the image is a solid blue color. Overlaid on this is a complex, abstract pattern of magenta lines and shapes. The pattern consists of various geometric elements: straight lines of different lengths and orientations, semi-circular arcs, and solid magenta triangles (one pointing up in the top right and one pointing down in the bottom left). These elements are interconnected to form a series of irregular, angular shapes that resemble a stylized, abstract architectural or biological structure.

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The Baculum in Microtine Rodents

Sydney Anderson



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Title: The Baculum in Microtine Rodents

Author: Sydney Anderson

Release date: November 15, 2011 [eBook #38021]

Language: English

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Volume 12, No. 3, pp. 181-216, 49 figs.
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Editors: E. Raymond Hall, Chairman, Henry S. Fitch, Robert W. Wilson

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UNIVERSITY OF KANSAS
Lawrence, Kansas

PRINTED IN
THE STATE PRINTING PLANT
TOPEKA, KANSAS
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28-774

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INTRODUCTION

Didier (1943, 1954) has described the bacula of several Old World microtines, and other rodents. Argyropulo studied (1933a, 1933b) five species of Cricetinae and *Microtus socialis*. Ognev (1950) illustrated numerous species of Eurasian microtines. Hamilton (1946) figured and described the baculum of 11 species of North American microtines. Hibbard and Rinker (1942, 1943) figured the baculum of *Synaptomys cooperi paludis* and of *Microtus ochrogaster taylori*. Dearden (1958) studied the baculum in two Asiatic species of *Lagurus*, in six subspecies of *Lagurus curtatus* of North America, and in six other species of microtines of other genera.

The baculum can be preserved easily with standard study skins, and is potentially useful in interpreting relationships on any taxonomic level, and especially in determining the relationships of species within a genus, if used together with other structures.

The anatomical orientation of the baculum needs comment because some confusion exists in the literature, especially concerning the use of the terms ventral and dorsal. The urethra lies on the anatomically ventral side of the penis, and of the baculum. In the center of the penis lies a single corpus cavernosum penis, shown in cross section proximal to the baculum in Figure 1c. Dorsally an artery, thinner walled than the ventral urethra, ends in a somewhat reticulate sinus surrounding primarily the middle part of the baculum within the bulbous glans penis. The corpus cavernosum penis (the structure has no median septum, at least distally) terminates with the baculum and is closely knit to it. The site of this bond is evident in the tuberosities and sculpturing of the base of the baculum.

The part of the penis enclosing the baculum, when not erect, is folded back as shown in Figures 1a and 1b. As a result the anatomically ventral surface faces upwards, or at least posterodorsally. The use of the term ventral in this

account refers to the anatomically ventral side, that is to say to the side of the baculum facing the urethra.

The baculum in microtines consists of an elongate stalk, having a laterally, and to a lesser extent dorsoventrally, expanded base and an attenuate distal shaft. Usually, three digitate processes of cartilaginous material in which additional ossifications may occur arise from the terminus of the shaft. The proportions and curvature of the stalk vary as do the proportions of the terminal ossifications to each other and to the stalk. In some species one or more of the digital processes are frequently completely unossified.

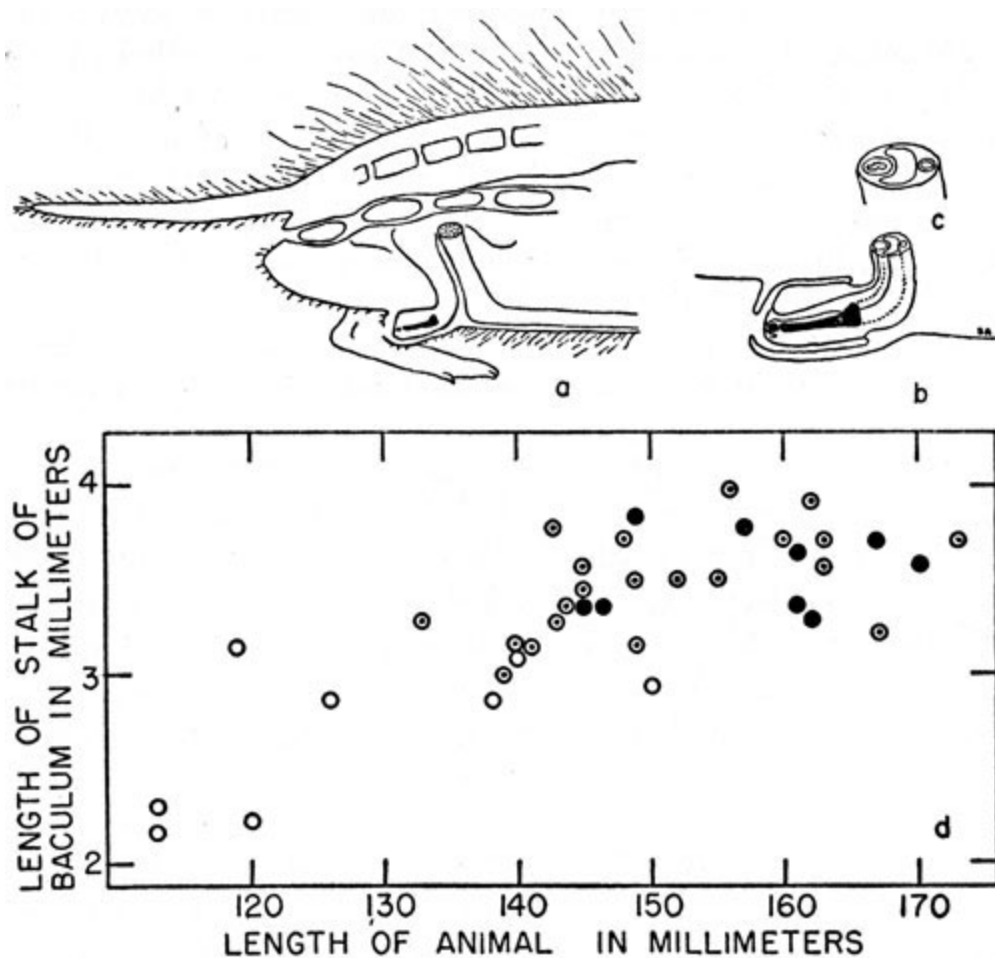


FIGURE 1. The baculum in *Microtus ochrogaster*—orientation and variation with age. *a*. Diagram of a sagittal section of the posterior half of a vole, natural size. The penis, containing the baculum (in black), extends ventrally from a point posterior to the pubic symphysis (stippled), along the body wall, and bends posteriorly at the distal end. *b*. Distal end of penis ($\times 2$)

showing baculum (in black), the urethra (solid lines) adjacent to the baculum, and the corpus cavernosum (broken lines) proximal to the baculum. *c.* Oblique view of the cross section of penis ($\times 4$) shown in Figure 1 *b*. The thick-walled urethra lies ventral to the curved corpus cavernosum. A thinner-walled blood-vessel lies dorsal to the corpus cavernosum. The anatomically ventral side of the baculum, in the normal non-erect penis shown, is seen to face dorsally. *d.* Graph showing the relationship between size of baculum, size of animal, and development of digital ossifications. Circles show presence of ossification in stalk only; circles enclosing dots indicate presence of secondary ossification in median process also; large dots indicate the addition of tertiary ossification in one or both of the lateral digitate processes.

Preserved specimens of *Microtus arvalis*, *Microtus agrestis*, *Microtus orcadensis*, *Microtus nivalis*, *Microtus guentheri*, *Microtus subterraneus*, *Clethrionomys glareolus*, and *Ellobius lutescens* were provided by Prof. Robert Matthey of Lausanne, Switzerland. J. Knox Jones, Jr. carefully saved the bacula with specimens of *Microtus fortis* and *Clethrionomys rufocanus* from Korea. Dr. W. B. Quay, Department of Zoology, University of California, supplied specimens of *Synaptomys cooperi*, *Phenacomys intermedius*, and *Microtus oregoni*. Dr. Franklin Sturges and Mr. John W. Goertz, Museum of Natural History, Oregon State College, Corvallis, have provided specimens including bacula of *Clethrionomys occidentalis*, *Microtus oregoni*, and *Microtus townsendii*. Dr. Randolph L. Peterson and Mr. Bristol Foster, Royal Ontario Museum of Zoology, Toronto, Canada, provided specimens of *Phenacomys intermedius*. Dr. J. N. Layne, University of Florida, Gainesville, Florida, presented me with a baculum of *Microtus parvulus*.

I am indebted to all of these persons for their aid, and to various collectors for the Museum of Natural History, who preserved bacula with specimens. Many of these specimens were obtained through the assistance of the University of Kansas Endowment Association and the National Science Foundation.

METHODS

Bacula were obtained from fresh specimens, specimens preserved in alcohol or formalin, and dried study skins. The processing of bacula has been discussed by Hamilton (1946), Friley (1947), White (1951), and Dearden (1958). The methods used to preserve bacula for my study differed some from any of those reported. The terminal part of each penis including the baculum imbedded in the glans penis was removed in its entirety and placed in a vial. The catalogue number was kept with each specimen at all times. A two per cent solution of potassium hydroxide was added. All specimens were examined at least once a day. If tissues other than the glans penis were present they were removed with forceps when softened usually at the end of one day. Several drops of Alizarin red-S stain in a saturated alcoholic solution were added to the 3 to 5 ccs. of KOH solution in each vial. Solutions were replaced if they became turbid enough to obstruct observation of the clearing penis. After one day the solution containing stain was removed and replaced with two per cent KOH solution without stain. When the glans became sufficiently cleared that the stained baculum could be seen easily, the solution was replaced by glycerin in which clearing was completed. The time required for the entire process varied from one day to more than two weeks depending on the size of the specimen and on its condition. Fresh specimens clear more rapidly than dried specimens, and those that are dried more rapidly than those that are preserved. A three or four per cent solution of hydroxide will hasten the process, but more frequent observation is required to prevent excessive maceration.

Specimens were then examined in a shallow dish containing glycerin under a binocular microscope. The baculum can be viewed from any desired direction. The method described above leaves the baculum intact within the glans penis; therefore its orientation can be determined relative to the thick walled urethra and the thin walled dorsal artery that extends onto the dorsal

side of the baculum. The ventral curvature of the penis proximal to the baculum, and the distal extension, characteristic of most species, of the dorsal border of the glans (both shown in Figure 1) are other features aiding in correctly orienting cleared specimens. The digitate processes are not so often injured, lost, or displaced when the method described above is used as they are when the penis is dissected. Specimens were stored in glycerin in glass shell vials having polyethylene stoppers. A small card bearing the name, number, locality, and other data was placed in each vial. A specimen thus enclosed can be kept indefinitely, or removed and mounted in balsam as described by White (1951:631) or in plastic as described by Dearden (1958:541) and thus stored in the vial containing the skull of the specimen.

Drawings were made on millimeter ruled paper while the baculum was viewed under a binocular microscope with a square ruled eyepiece.

Unless otherwise noted all specimens listed are in the University of Kansas Museum of Natural History. Catalogue numbers are cited. Measurements are accurate to within less than one-tenth of a millimeter. Proportions as stated in the text are approximations, accurate to within one-twelfth (8.33 per cent). The range of variation is unknown for some species. Mention is made if maturity is known or suspected to differ in specimens being compared.

The development of the baculum has been studied by Callery (1951) in *Mesocricetus auratus* and by Ruth (1934) in the laboratory rat. In the rat (*Rattus norvegicus*) the bone is of endoblastemal origin being laid down by a condensation of undifferentiated mesenchymal cells. At the distal end of the bone dense fibrous tissue is then differentiated and at the proximal end hyaline cartilage. Growth is by substitution at the proximal end and by subperiosteal lamellation circumferentially. A marrow cavity is formed by resorption. In the baculum of the hamster the primary center of ossification is in the stalk, and is present at the age of three days; the secondary centers are in lateral processes and are present at 80 days and enlarge subsequently. A tertiary center, in each median process, may or may not develop later. Maximum development of the baculum is reached late in the reproductive life of the hamster.

The early ossification of the baculum noted in the rat and the hamster occurs in *Microtus* also. A specimen of *Microtus montanus fusus* (76831, from 5 mi. N, 26 mi. W Saguache, 9600 ft., Saguache County, Colorado) only 74 mm. in total length and weighing only 6.6 grams, had a slender ossified baculum having enlarged ends. This vole was one-half of the average length and less than one-fifth of the average weight of an adult, and of approximately the size at which weaning takes place.

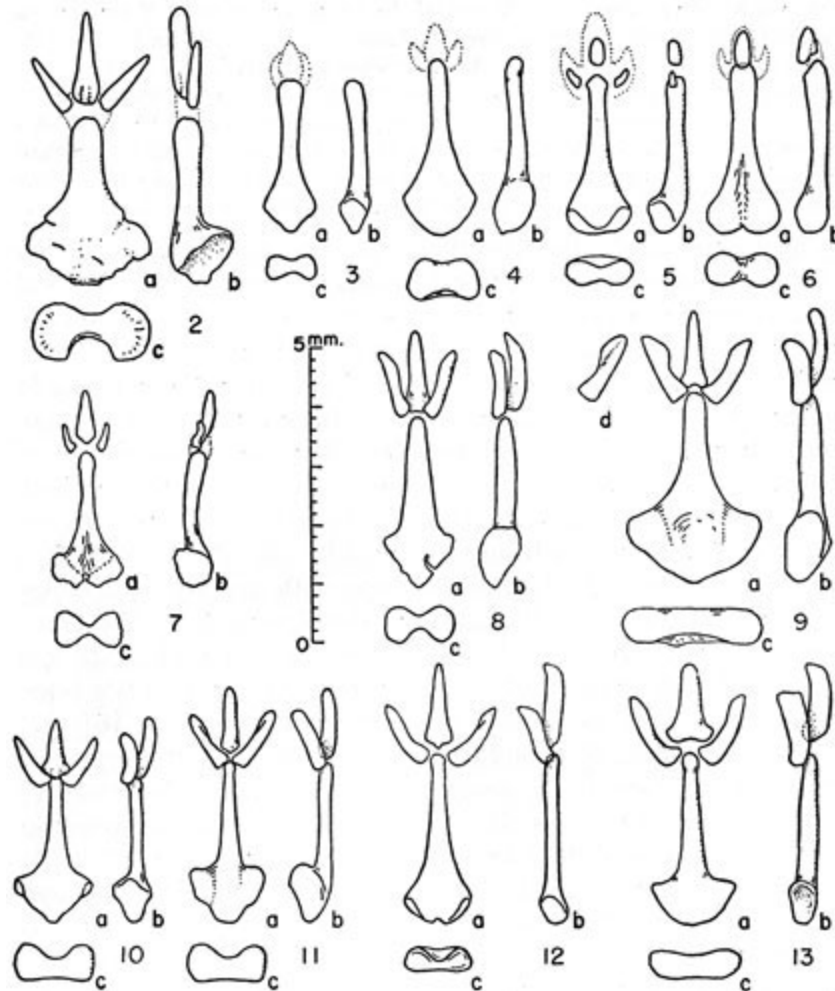
The development of the baculum in *Microtus ochrogaster* was studied in 32 specimens of various ages. The specimens (between Nos. 74994 and 75074) were collected between August 15 and September 4, 1957, at localities on the Great Plains. These specimens were from breeding populations, as evidenced by pregnancy of females and by large size of testes of males. The length and width of the stalk of the baculum, the presence of digital ossifications, the total length of the animal, and the size of the testes were noted. Variability in length of testes is greatest when voles are from 140 to 150 mm. in total length. Sexual maturity is reached rather abruptly when the total length of most individuals is 140 to 150 millimeters. If the baculum likewise underwent more rapid growth at the onset of sexual maturity, greater variability should be evident in the length of the baculum of voles 140 to 150 mm. in total length than in bacula of voles of other sizes. This was the case (see Figure 1d). The baculum does not, however, suddenly reach its maximum maturity.

The primary ossification is in the stalk. The secondary ossification is in the median process except in *Lagurus* (Dearden, 1958:551) and some individuals of *Neofiber* (see account on page 258). Tertiary centers of ossification are in the lateral processes. The primary ossification is present at an early age and subsequently increases in size and solidity. The secondary and tertiary ossifications are progressively more common in older voles. The increase in degree of ossification of all parts continues after sexual maturity is reached. Individual variation and variation with age in the baculum of *Microtus pennsylvanicus* have been illustrated by Hamilton (1946:380). Figures 14, 15, and 17 illustrate variation with size, which is correlated with age, and also illustrate individual variation. The three bacula are from adult voles having testes that measured 15, 16 and 16 mm. in length, respectively. Each vole was trapped in late June. The total lengths in millimeters of the three voles are 172, 167, and 181; weights are

55, 52.4, and 65.5 grams. I judge that the greater size of the stalk and the better developed base shown in Figure 17 than in Figure 15 are illustrative of age variation; the difference in the size of the lateral digitate processes is, in this case, attributable to individual variation. Differences in the distal end of the baculum in Figures 42 and 43, show individual variation also. Figures 35 and 36 represent two different subspecies; different individuals of *M. mexicanus mogollonensis*, however, exhibit individual variation of the same degree.

Hall and Cockrum (1953) list 44 species of microtines in North America. At least twelve of these are insular or local forms perhaps derived from some other species; for example *Microtus coronarius*, an insular form derived from *Microtus longicaudus*; *Microtus propectus*, considered by Chamberlain (1954:587) and by Wheeler (1956:176) as a subspecies of *Microtus pennsylvanicus*; and *Microtus ludovicianus*, a close relative of *Microtus ochrogaster*.

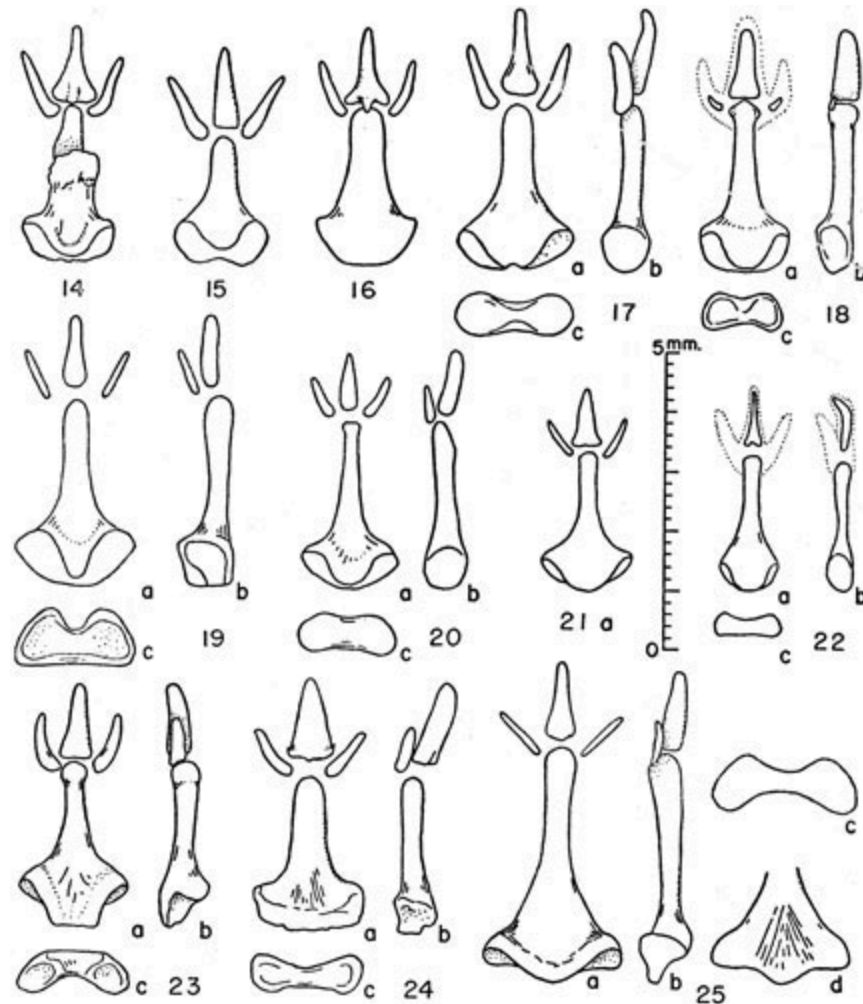
All North American genera have been studied. Of the genus *Microtus* in North America, all subgenera but *Orthriomys* and all species but the following nine, have been studied: *M. (Orthriomys) umbrosus*, the insular *M. (Stenocranius) abbreviatus*, *M. (Microtus) breweri*, *M. (Microtus) nesophilus*, *M.*



FIGURES 2-13. Bacula of microtines. Unless indicated otherwise views are (a) of the dorsum, (b) the right side, and (c) the proximal end with the dorsal surface upward. Exact localities are given in accounts of species concerned.

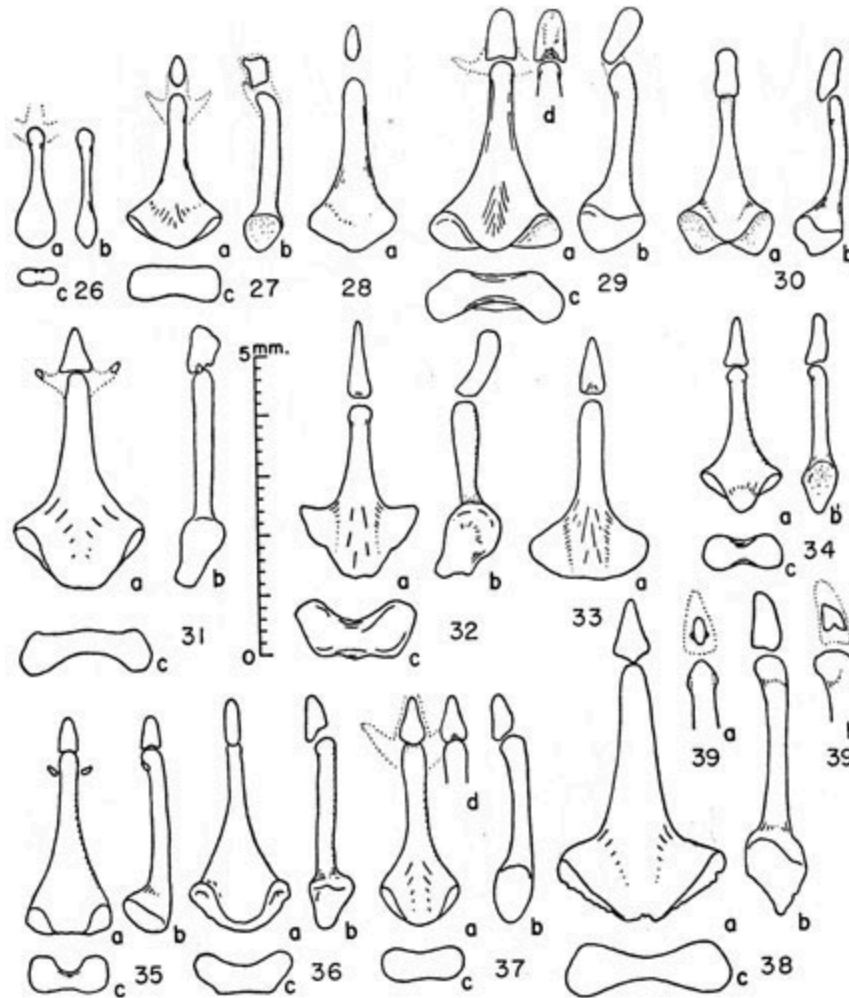
2. *Lemmus trimucronatus*, 50678, Point Barrow, Alaska.
3. *Dicrostonyx groenlandicus*, 50539, Porcupine Lake, Brooks Range, Alaska.
4. *Dicrostonyx groenlandicus*, 52524, Point Barrow, Alaska.
5. *Synaptomys cooperi saturatus*, WBQ 3-C-454, 3 mi. S Demotte, Indiana.
6. *Synaptomys cooperi paludis*, 13716, Meade County State Park, Kansas.
7. *Phenacomys intermedius celatus*, SA 2044, Quebec.

8. *Phenacomys intermedius intermedius*, WBQ 3-C-309, 5.4 mi. S Moran, Teton Co., Wyoming.
9. *Clethrionomys rufocanus*, 60438, 1 mi. NW Oho-ri, Korea, (d) ventral view.
10. *Clethrionomys gapperi*, 42108, 31 mi. N Pinedale, Wyoming.
11. *Clethrionomys rutilus*, 42865, 5 mi. NNE Gulkana, Alaska.
12. *Clethrionomys occidentalis*, FWS 30, Mary's Peak, Benton Co., Oregon.
13. *Clethrionomys glareolus*, 67100, Zermatt, Valais, Switzerland.



FIGURES 14-25. Bacula of *Microtus*. Unless indicated otherwise views are (a) of the dorsum, (b) the right side, and (c) the proximal end with dorsal surface upward.

14. *M. pennsylvanicus*, 42439, 1 mi. S, 2 mi. E Eagle Nest, Colfax Co., New Mexico; abnormality perhaps owing to injury; dorsal view.
15. *M. pennsylvanicus*, 42306, 5 mi. N, 26 mi. W Saguache, Colorado; dorsal view.
16. *M. pennsylvanicus*, 43043, 20 mi. NE Anchorage, Alaska, ventral view.
17. *M. pennsylvanicus*, 42430, 1 mi. S, 2 mi. E Eagle Nest, New Mexico.
18. *M. agrestis*, 67102, Gryon, Switzerland.
19. *M. montanus amosus*, 62241, 1/2 mi. E Soldier Summit, Wasatch Co., Utah.
20. *M. montanus nanus*, 57470, 2 mi. N, 2 mi. W Pocatello, Idaho.
21. *M. montanus fusus*, 42307, 5 mi. N, 26 mi. W Saguache, Colorado.
22. *M. arvalis*, 67101, Vidy, Switzerland, possibly not mature.
23. *M. guentheri*, 67104, Palestine.
24. *M. orcadensis*, 67106, Orkney Islands, orientation uncertain.
25. *M. fortis*, 63841, Chipori, Korea, (*d*) ventral view.



FIGURES 26-39. Bacula of microtines. Unless indicated otherwise views are (a) of the dorsum, (b) the right side, and (c) the proximal end with the dorsal surface upward.

26. *M. (Pitymys) fatioi*, 67103, Zermatt, Switzerland, immature.

27. *M. (Pitymys) pinetorum*, 76834, 2 mi. N Baldwin, Douglas Co., Kansas.

28. *M. (Pitymys) pinetorum*, 68545, 1 mi. NE Pleasant Grove, Kansas.

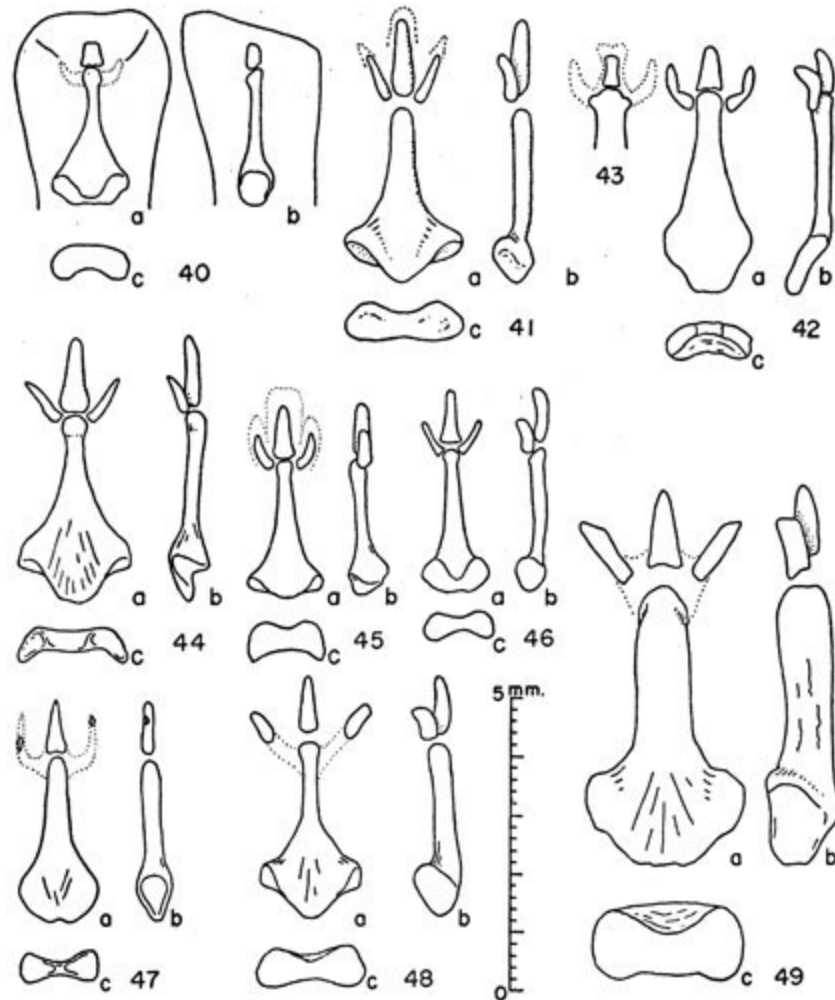
29. *M. (Pitymys) quasiator*, 30709, Teocelo, Veracruz, (d) ventral view.

30. *M. (Pitymys) quasiator*, 19878, 5 km. N Jalapa, Veracruz.

31. *M. (Pedomys) ochrogaster*, 75036, 1 mi. N, 2 mi. E Oberlin, Kansas.

32. *M. (Stenocranius) miurus*, 51152, Lake Schrader, Brooks Range, Alaska.

33. *M. (Stenocranius) miurus*, 51169, Lake Schrader, Brooks Range, Alaska.
34. *M. (Stenocranius) gregalis*, 8059, "Eastern Europe."
35. *M. mexicanus mexicanus*, 63094, Valle de Bravo, Estado de México, México.
36. *M. mexicanus mogollonensis*, 63298, Mt. Taylor, Valencia Co., New Mexico.
37. *M. californicus*, 76828, 1 mi. NE Berkeley, California; (d) ventral view.
38. *M. (Arvicola) richardsoni*, 42454, 31 mi. N Pinedale, Sublette Co., Wyoming.
39. *M. richardsoni*, 37903, 23-1/2 mi. S, 5 mi. W Lander, Wyoming; distal end.



FIGURES 40-49. Bacula of microtines. Unless indicated otherwise views are (a) of the dorsum, (b) the right side, and (c) the proximal end with the dorsal surface upward.

40. *Microtus (Pitymys) parvulus*, UF 1508, 1 mi. W Micanopy, Florida.
41. *Microtus townsendii*, 79186, Sec. 33, T. 11S, R. 5W, Benton Co., Oregon.
42. *Microtus (Herpetomys) guatemalensis*, 65895, 2 mi. S San Juan Ixcoy, Guatemala.
43. *M. guatemalensis*, 65921, 10 mi. E, 4 mi. S Totonicapan, Guatemala, dorsal view of tip.
44. *Microtus oeconomus*, 43048, Kelsall Lake, British Columbia.
45. *Microtus (Chilotus) oregoni*, WBQ 3-C-248, 5 mi. N Orick, California.
46. *Lagurus (Lemmiscus) curtatus*, 26053, 9 mi. S Robertson, Uinta Co., Wyoming.
47. *Microtus (Chionomys) nivalis*, 65127, Wetterstein, Germany, orientation uncertain.
48. *Microtus (Chionomys) longicaudus*, 50253, Crane Flat, Mariposa Co., California.
49. *Neofiber alleni*, 27268, 2 mi. S Gainesville, Florida, orientation uncertain.

(*Microtus*) *provectus* (the last three are probably insular derivatives of *M. pennsylvanicus*), *M. (Microtus) fulviventer* (perhaps derived from the same stock as *Microtus mexicanus*), *M. (Microtus) xanthognathus* (perhaps related to *Microtus chrotorrhinus*), *M. (Microtus) coronarius*, and *M. (Pedomys) ludovicianus*.

SPECIES OF WHICH BACULA WERE EXAMINED

Subfamily: Microtinae	Number of Specimens
Tribe: Lemmi	
<i>Dicrostonyx groenlandicus</i> (Traill)	4
<i>Lemmus trimucronatus</i> (Richardson)	6
<i>Synaptomys cooperi</i> Baird	5

Tribe: Microti

Genus: *Clethrionomys* Tilesius, 1850

<i>Clethrionomys rutilus</i> Pallas	4
<i>Clethrionomys gapperi</i> (Vigors)	9
<i>Clethrionomys occidentalis</i> (Merriam)	1
<i>Clethrionomys glareolus</i> Schreber	1
<i>Clethrionomys rufocanus</i> Sundevall	1

Genus: *Phenacomys* Merriam, 1897

<i>Phenacomys intermedius</i> Merriam	5
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Genus: *Ondatra* Link, 1795

<i>Ondatra zibethicus</i> (Linnaeus)	1
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Genus: *Microtus* Schrank, 1798

(<i>Herpetomys</i>) <i>guatemalensis</i> Merriam	3
(<i>Arvicola</i>) <i>richardsoni</i> (DeKay)	2
(<i>Chilotus</i>) <i>oregoni</i> (Bachman)	3
(<i>Stenocranius</i>) <i>gregalis</i> (Pallas)	1
(<i>Stenocranius</i>) <i>miurus</i> Osgood	9
(<i>Chionomys</i>) <i>longicaudus</i> (Merriam)	6
(<i>Chionomys</i>) <i>nivalis</i> Martins	2
(<i>Microtus</i>) <i>arvalis</i> (Pallas)	1
(<i>Microtus</i>) <i>orcadensis</i> Millais	1
(<i>Microtus</i>) <i>guentheri</i> Danford and Alston	1
(<i>Microtus</i>) <i>fortis</i> Büchner	2
(<i>Microtus</i>) <i>montanus</i> (Peale)	15
(<i>Microtus</i>) <i>townsendii</i> (Bachman)	3
(<i>Microtus</i>) <i>oeconomus</i> (Pallas)	10
(<i>Microtus</i>) <i>mexicanus</i> (Saussure)	13
(<i>Microtus</i>) <i>californicus</i> (Peale)	2
(<i>Microtus</i>) <i>pennsylvanicus</i> (Ord)	13
(<i>Microtus</i>) <i>agrestis</i> (Linnaeus)	1
(<i>Pedomys</i>) <i>ochrogaster</i> (Wagner)	41

<i>(Pitymys) pinetorum</i> (LeConte)	2
<i>(Pitymys) parvulus</i> (Howell)	1
<i>(Pitymys) quasiater</i> (Coues)	5
<i>(Pitymys) fatioi</i> Mottaz	1
Genus: <i>Neofiber</i> True, 1884	
<i>Neofiber alleni</i> True	2
Genus: <i>Lagurus</i> Gloger, 1841	
<i>Lagurus curtatus</i> (Cope)	7
Total number examined	184

ACCOUNTS OF SPECIES

Dicrostonyx groenlandicus (Traill)

Figs. 3 and 4

Baculum: stalk elongate, greatest length (3.1 mm.) $2 \frac{1}{5}$ to $2 \frac{1}{2}$ times greatest breadth, and $4 \frac{1}{2}$ times greatest depth; digitate processes usually cartilaginous, occasionally lateral processes partly ossified; basal tuberosities weakly to moderately developed, medially confluent; posterior profile in dorsal view rounded with rounded posterior apex or shallow notch; dorsal concavity in end-view shallower and not so wide as ventral concavity; median constriction approximately $\frac{2}{3}$ greatest depth; ventral part of base in end-view wider than dorsal part; shaft straight or slightly curved; base of stalk placed dorsally relative to axis of shaft; stalk spatulate, sometimes with distal enlargement; at mid-point stalk wider than high; lateral profile in dorsal view sloping gradually without abrupt curvature anterior to point of greatest width.

The baculum of *Dicrostonyx torquatus* figured by Ognev (1948:476) agrees with that of *D. groenlandicus* in shape of stalk, and in lateral digitate processes that are small relative to size of median process; but differs in more elongate, terminally enlarged, bulbar shape of median process. None of my specimens showed ossification in the lateral processes, observed by Hamilton (1946:381) in *Dicrostonyx rubricatus richardsoni* [= *D. groenlandicus richardsoni*]. In all of my specimens the cartilaginous median process was larger than that figured by Hamilton, or by Dearden (1958:542).

Specimens examined: Four from; Point Barrow, Alaska, 52524 (Barrow Village), 67264 (died in captivity); Brooks Range, Alaska, 50536 (Wahoo

Lake, 69°08', 146°58'), 50539 (Porcupine Lake, 68°51'57", 146°29'50", 3140 ft.).

Lemmus trimucronatus (Richardson)

Fig. 2

Baculum: Stalk heavy, broad, greatest length (2.8 mm.) in mature individuals (Fig. 2) as little as $1\frac{1}{3}$ times greatest breadth, greatest length no less than $2\frac{2}{3}$ times greatest depth of base; three ossified processes, median one from as long as to $\frac{1}{2}$ longer than the lateral processes, and approximately $\frac{2}{3}$ wider and twice as deep as lateral processes; length of median process almost $3\frac{1}{2}$ times its breadth, approximately $\frac{1}{2}$ length of stalk; basal fossae broadly confluent; posterior profile in dorsal view evenly rounded; in end-view ventral concavity deeper than dorsal concavity, constriction as little as $\frac{1}{2}$ greatest depth in mature specimens; shaft straight, bluntly rounded, or slightly decurved and laterally inflated terminally; lateral profile in dorsal view a gradual slope from widest point of stalk anteriorly onto shaft; in younger individuals stalk slenderer, otherwise as described above.

Five specimens examined by me differ from one figured and described by Hamilton (1946:379) in that stalk is better developed, larger relative to size of processes, length of stalk in my specimen (Fig. 2) 2.8 as opposed to 2.1 mm. in Hamilton's specimen; median process shorter, 1.5 as opposed to 1.8 mm., proximal end rounded rather than concave, not partially enclosing tip of shaft; proportion of and relative sizes of median and lateral processes approximately same as in Hamilton's *Lemmus helvolus* [= *Lemmus trimucronatus helvolus*]. A specimen figured by Dearden (1958:542) has a basally trilobed median process.

The baculum of the Asiatic *Lemmus lemmus* figured by Ognev (1948:413) agrees with my specimens in the ossification of three processes, the relative sizes of these processes to each other and to the stalk, the well-developed base of the stalk and heavy bluntly rounded shaft; the baculum of *Lemmus lemmus* differs in greater anterolateral extent of basal tuberosities, in

proximal notch seemingly separating these tuberosities, and in median process being slenderer.

Specimens examined: Five, of two subspecies; *Lemmus trimucronatus alascensis*, Point Barrow, Alaska, numbers 50591, 50678, 50731, 50758; *Lemmus trimucronatus subarcticus*, Wahoo Lake, 69°08', 146°58', 2350 ft., Brooks Range, Alaska, 50948.

Synaptomys cooperi Baird

Figs. 5 and 6

Baculum: Stalk elongate, greatest length (2.7 to 2.8 mm.) $2 \frac{1}{3}$ to $2 \frac{1}{2}$ times greatest breadth, 4 to 5 times greatest depth; three processes ossified or lateral processes unossified, ossifications relatively small (in 78380, median ossification less than $\frac{1}{4}$ as large as lateral ossifications although median cartilaginous process is larger), length of median process $\frac{1}{5}$ to $\frac{1}{6}$ of length of stalk, cartilaginous part of median process larger; posterior profile in dorsal view convex throughout or bilobate; tuberosities moderately developed, deflected dorsal to axis of shaft; in end-view medial construction $\frac{3}{5}$ greatest depth of tuberosities; shaft tapered from point of greatest width, slightly inflated terminally.

The specimen (KU 13716) figured by Hibbard and Rinker (1942:29) has been restudied. It was first cleared and stained to soften the dry cartilage binding the digital processes together and to differentiate bone and cartilage. The lateral processes are small and cartilaginous (Fig. 6) and seem intact. The differences between this specimen and others examined by Hamilton (1946:381), Dearden (1958:542), and myself, namely the relatively larger median ossification, the absence of ossification in lateral processes, and the distinctly bilobate base and larger size, may represent geographic differences, or individual variation. The proportions of length, width, and depth of the stalk, and the appearance in lateral view do not differ greatly from others examined by Hamilton, by Dearden (1958:546), and by me.

Specimens examined: Five, representing four subspecies; *S. cooperi gossii*, 6 mi. N Midway, Holt Co., Nebraska 78379, 78380; *S. cooperi relictus*, 5 mi. N, 2 mi. W Parks, Dundy Co., Nebraska, 72601 (immature); *S. cooperi saturatus*, 3 mi. S Demotte, Jasper Co., Indiana, 3-C-454, collection of W. B. Quay; *S. cooperi paludis*, Meade County State Park, Kansas, 13716.

***Clethrionomys rutilus* Pallas**

Fig. 11

Baculum: Stalk elongate, and proximally enlarged, greatest length (2.7 mm.) 2 times greatest breadth; less than 4 times greatest depth; three well-developed ossified processes; length of stalk $2\frac{1}{3}$ times length of median process; median process with basal (and ventral) protuberance and lateral lobes, arched in dorsoventral plane; lateral processes as large as median process, flattened distally, having ventromedial vane on distal half; basal tuberosities of stalk well developed, medially confluent; posterior profile in dorsal view trilobate or convex throughout with rounded posterior apex; dorsal concavity well developed, ventral surface but slightly concave, medial constriction of base as little as $\frac{1}{2}$ greatest depth; shaft straight, slender, at mid-point of stalk but slightly wider than high; basal tuberosities largely dorsal to axis of shaft in lateral view; lateral profile in dorsal view with an abrupt curvature separating the gently sloping sides of the shaft from the basal part at its greatest breadth.

The specimen of *Clethrionomys rutilus* figured by Ognev (1950:120) is essentially like the North American specimens examined by me in the relative sizes of the ossifications and the general shape of the stalk.

Specimens examined: Four, of one subspecies; *C. r. dawsoni*, west bank Gakona River, 1700 ft., 5 mi. NNE Gulkana, Alaska, 42865, 42866; SW end Dezadeash Lake, 2400 ft., Yukon Territory, 42910, 42921.

***Clethrionomys gapperi* (Vigors)**

Fig. 10

Baculum: Stalk elongate, greatest length (2.8 mm.) $1\frac{3}{4}$ times greatest breadth, and $3\frac{3}{4}$ times greatest depth; proximally enlarged, greatest depth $\frac{1}{2}$ greatest breadth; three well-developed ossified processes; length of stalk $2\frac{1}{3}$ times length of median process; median process arched in dorsoventral plane, with basiventral protuberance or spine and lateral lobes; lateral processes as large as median process, flattened distally, arched; basal tuberosities of stalk well developed, medially confluent; posterior profile in dorsal view trilobate or convex throughout with a rounded posterior apex; dorsal concavity well developed, ventral surface but slightly concave, or in some cases slightly convex; medial constriction of base $\frac{3}{5}$ greatest depth; shaft straight, slender, at mid-point of stalk twice as wide as high; basal tuberosities dorsally placed relative to axis of shaft; lateral profile in dorsal view abruptly curved anterior to point of greatest width; slender stalk distinct from angular enlarged base.

The most noticeable difference between the baculum of *C. rutilus* and *C. gapperi* is size. The proportions of the four ossifications are approximately the same. Ventral vanes on the lateral processes are not developed in *C. gapperi*. *C. gapperi* and *C. rutilus* are more nearly alike in their bacula than any other two species of *Clethrionomys* examined. *Clethrionomys occidentalis*, the other New World species, is also much like *C. gapperi* and *C. rutilus*. The differences are of a magnitude comparable to those between the bacula in subspecies of *Microtus montanus* (Figs. 19-21) for example, or in subspecies of *Lagurus curtatus* (Dearden, 1958:542).

Specimens examined: Nine, of two subspecies; *Clethrionomys gapperi athabasca*, British Columbia, 42922 (Indian Creek, Mile Post 234 of Alaskan Highway), 64281 (West bank Racing River, 89 mi. W Muskwa), 64287 (North bank Tetsa River, 56 mi. W, 11 mi. S Muskwa), 64290 (44 mi. W, 9 mi. S Muskwa), 64310 (32 mi. W, 2 mi. S Muskwa); *Clethrionomys gapperi galei*, 31 mi. N Pinedale, Sublette Co., Wyoming, 42108; Grand Mesa, Delta Co., Colorado, 60014 and 60015 ($5\frac{1}{2}$ mi. E, 12 mi. S Collbran), 60022 (8 mi. E, $\frac{1}{2}$ mi. S Skyway).

Clethrionomys occidentalis (Merriam)

Fig. 12

Baculum: Stalk elongate, greatest length (2.8 mm.) $2\frac{1}{2}$ times greatest breadth, 6 times greatest depth; three well-developed ossified processes; median process larger than lateral processes, $\frac{1}{2}$ the length of stalk, curved, basally broad, ventrally keeled, trilobate posteriorly; lateral ossifications large, flattened distally, curved; posterior profile of stalk posteriorly slightly emarginate, thus bilobate in outline; in end-view dorsal concavity deeper than ventral, constriction less than $\frac{1}{2}$ greatest depth, tuberosities confluent, visible in dorsal view at each side; shaft slender, especially in depth, straight; at mid-point of stalk almost twice as wide as deep, slight terminal inflation.

The general proportions of the stalk and the relatively large, uniquely shaped processes, are characteristic of most specimens of the genus *Clethrionomys* examined.

Specimen examined: *C. occidentalis californicus*, one from Mary's Peak, Benton Co., Oregon, 30, F. W. Sturges' collection.

Clethrionomys glareolus Schreber

Fig. 13

Baculum: Stalk elongate, greatest length (2.9 mm.) twice the greatest breadth in the specimen examined, flattened proximally, greatest length almost 6 times greatest depth of base; three well-developed ossified processes; median process arched in a dorsoventral plane, with basal notch and lateral lobes; lateral processes as long as median process, bowed in dorsal view, flattened distally, with ventromedial vane; basal tuberosities of stalk weakly developed, medially confluent; posterior profile in dorsal view evenly rounded; in end-view dorsal concavity shallow in comparison to most species but deeper than ventral concavity, constriction $\frac{3}{4}$ greatest depth; shaft straight, at mid-point slightly wider than high, elongate, widest point of stalk less than $\frac{1}{4}$ of total length from proximal end, slight lateral

inflation at tip; lateral profile in dorsal view sloping at first abruptly and then gradually from widest point of stalk anteriorly onto shaft.

The specimen of *Clethrionomys glareolus* figured by Ognev (1950:31) in dorsal view as I interpret it, resembles my specimen in the rounded base; in the elongate, distally inflated shaft; in the initially abrupt slope of the lateral profile in dorsal view from the greatest width of stalk anteriorly; and in the presence of three well ossified processes. Ognev's specimen differs from mine in the median process being more elongate relative to its width, and rounded proximally, lacking lateral lobes and basal notch; in lateral processes being less curved; in the greater terminal inflation of the shaft; and in the closer approximation of the terminal processes to the shaft. The baculum of *Clethrionomys glareolus* as described and figured by Didier (1954:243-244) resembles my specimen in general proportions, but is more pointed proximally and more curved in dorsoventral plane. Didier states that the baculum is rather variable in form in this species, in different regions, but that a large number of specimens must be examined to assess the geographic nature of this variation.

Specimen examined: One from Zermatt, Valais, Switzerland, 67100.

***Clethrionomys rufocanus* Sundevall**

Fig. 9

Baculum: Base of stalk broad but relatively flattened dorsoventrally, greatest length (3.2 mm.) less than $1\frac{1}{2}$ greatest width, 4 times greatest depth; three well-developed ossified processes; median process arched in dorsoventral plane, having basal notch and lateral lobes; lateral processes as long as median process, flattened distally, with ventromedial vane; basal tuberosities of stalk weakly developed, medially confluent; posterior profile in dorsal view convex with rounded posterior apex; dorsal surface of base almost flat, ventral concavity broad and shallow; constriction $\frac{3}{4}$ greatest depth (not including an unusual irregularity on the ventral surface of the base); shaft straight, at mid-point of stalk distinctly wider than high, slender at distal end, widest point of stalk almost $\frac{1}{3}$ of total length from proximal

end, tip of shaft rounded; lateral profile in dorsal view gradually sloping from widest point anteriorly onto shaft.

The specimen of *Clethrionomys rufocanus* figured by Ognev (1950:97) resembles my specimen in the presence of three well ossified processes. Ognev's specimen differs however in the lack of a proximal notch on the median process, the lesser proportion of the stalk included in the basal enlargement, the more posterior position of the point of greatest width, and the presence of a concavity in the posterior profile of the stalk in dorsal view. These differences in the stalk may be owing to a difference in age (my specimen perhaps being older).

Specimen examined: One from 1 mi. NW Oho-ri, 6 M., Korea, 60438.

Phenacomys intermedius Merriam

Figs. 7 and 8

Baculum: Stalk slender, greatest length (2.9 mm.) $2\frac{1}{4}$ to $2\frac{1}{2}$ times greatest breadth, 4 times greatest depth; three well-developed ossified processes, median one almost $\frac{1}{2}$ length of stalk, curved, broad basally and slightly larger in all dimensions than either lateral process; lateral processes flattened distally, curved; base of stalk well developed, basal tuberosities medially confluent or separated by medial emargination, posterolateral faces flattened or rough; emarginations in the four adults examined; posterior profile in dorsal view bluntly pointed or flattened except for emargination posteriorly, abruptly curved at point of greatest width; shaft arising broadly from distal side of base of stalk; in end-view hour-glass shaped, medial constriction pronounced, both dorsal and ventral concavities deep; shaft having relatively straight but distally convergent sides; at mid-point of stalk, 1 to $1\frac{1}{2}$ times as wide as deep; tip bluntly rounded, or slightly inflated.

The specimens from Quebec differ from the one from Wyoming in smaller size, relatively smaller lateral digital processes, larger more medial basal emargination, and slender shafts. The baculum of *Phenacomys intermedius* differs much from that of *Phenacomys longicaudus*, described by Hamilton

(1946:381) and by Dearden (1958:547). Dearden states that the three bacula examined by him of *Phenacomys longicaudus* differ markedly from the specimen described by Hamilton. It seems to me that in major features the resemblance is greater between the specimens of *Phenacomys longicaudus* examined by these two authors than between their specimens and specimens of other microtines, including *Phenacomys intermedius*. Neither Hamilton nor Dearden record the exact localities of capture, the collections in which the specimens are deposited, or the catalogue numbers of specimens. Consequently verification of identifications and observations is difficult.

Specimens examined: Five, of two subspecies; *P. intermedius intermedius*, 5.4 mi. S Moran, Teton Co., Wyoming, 3-C-309, collection of W. B. Quay; *P. intermedius celatus*, four (including one immature specimen) from Authiernord, Abitibi-ouest Co., Quebec, specimens in collection of Bristol Foster designated by numbers 2041-2044 of S. Anderson's field catalogue. Smith and Foster (1957:107) were of the view that *Phenacomys ungava* (including the above specimens from Quebec) may be specifically distinct from *Phenacomys intermedius*.

Ondatra zibethicus (Linnaeus)

Not figured

Baculum: In the single specimen examined, less mature than that figured by Hamilton (1946:384), the digitate processes are cartilaginous, the basal tuberosities are less well developed, and the shaft is slenderer throughout. The cartilaginous processes are of the same proportions as ossified processes in the figure mentioned. The shaft is also convex ventrally in lateral profile. The view of the side here considered to be anatomically the ventral side (adjacent to the urethra) is labelled dorsal view in Hamilton's specimen.

Specimen examined: One, from Reserve, Brown Co., Kansas, 72405.

***Microtus (Herpetomys) guatemalensis* Merriam**

Figs. 42 and 43

Baculum: Stalk moderately elongate, greatest length (3.5 mm.) $2\frac{1}{3}$ times greatest breadth, spatulate, flattened throughout, greatest thickness $\frac{1}{3}$ millimeter; three ossified processes; median process having three cornered base, curved dorsally, wider than high, $\frac{1}{4}$ to $\frac{1}{5}$ greatest length of stalk; each lateral process bent at middle, as long as median process, compressed laterally; base of stalk curved dorsally, tuberosities marginal, hence narrow, lateral excavations of tuberos margin not confluent medially; in end-view ventral concavity broad, no dorsal concavity, medial constriction but slightly less than greatest thickness (not depth); shaft wider than high throughout, at mid-point more than 3 times as wide as high; tip of shaft slightly inflated both laterally and dorsoventrally; lateral profile gradually sloping anteriorly from widest point of stalk.

Specimen number 65921 (Fig. 43) differs from number 65895 (Fig. 42) described above. Terminus of shaft of number 65921 has lateral lobes from which arise lateral cartilaginous processes; median terminal ossification irregular in shape, smaller, imbedded in terminally bilobate cartilage. In the spatulate flattened stalk these two specimens are much alike. An immature specimen, number 65908, is smaller (length of stalk 2.6 mm.) also flattened and spatulate, has the terminal processes cartilaginous, the lateral processes bent medially, and proportions as in the adult.

The baculum shows no noteworthy resemblance to that of any other species of North American *Microtus*; on the other hand the differences between *M. guatemalensis* and some other species are no greater than the differences between certain species included in the subgenus *Microtus*. The baculum neither strengthens nor weakens the case for subgeneric rank for *M. (Herpetomys) guatemalensis*.

Specimens examined: Three from Guatemala; 65895 (2 mi. S San Juan Ixcay), 65908, (3-1/2 mi. SW San Juan Ixcay), 65921 (10 mi. E, 4 mi. S Totonicapán).

***Microtus (Arvicola) richardsoni* (DeKay)**

Figs. 38 and 39

Baculum: Stalk broad, greatest length (3.7 to 4.3 mm.) $1\frac{1}{2}$ times greatest breadth, relatively flattened, greatest depth $\frac{1}{3}$ greatest breadth; single median ossified process, in smaller of two specimens this ossification incomplete and of unusual shape (Fig. 39); length of stalk 4 times length of median process; concavities of basal tuberosities medially confluent, constriction less than $\frac{1}{2}$ greatest depth; widest point of shaft less than $\frac{1}{4}$ length of shaft from posteriormost point; shaft wider than high except at distal end that is inflated dorsally and sometimes laterally; both ventral and dorsal concavities of base of stalk broad and moderately deep; posterior profile in dorsal view evenly rounded or having marginal notch.

In the absence of ossified lateral processes my two specimens differ from bacula of *Microtus (Arvicola) terrestris* figured by Didier (1943:79, 1954:245, 247, 248) and by Ognev (1950:591). The median process relative to the size of the shaft is smaller, and the shaft relative to its length is wider in *M. richardsoni* than in *M. terrestris*. The stalk of *M. (Arvicola) amphibius* figured by Didier is like that of *M. richardsoni* in its greater breadth and median notch on posterior border.

The relationship of the New World water rat, *M. richardsoni*, to the Old World water rats (genus *Arvicola* of some European authors) is uncertain. Miller (1896:66) placed all of them in the subgenus *Arvicola*. Subsequent authors, stressing differences in the teeth, have placed *M. richardsoni* in the subgenus *Aulacomys* of Rhoads. Zimmerman (1955) has shown that teeth in some *Arvicola* approach the more complex pattern of *M. richardsoni*. He argues also that *Arvicola* is generically distinct from *Microtus* on the grounds that the two groups have separate origins, *Arvicola* having descended from the genus *Mimomys* and *Microtus* from some other group of microtines. This argument also was advanced by Hinton (1926:47-48). Pending further studies of the possible polyphyletic origin of other subgenera of the genus *Microtus*, I refer both *M. richardsoni* and *M. terrestris* to the subgenus *Arvicola*.

The evidence afforded by the bacula available is not conclusive as to relations of Old World and New World water rats. No general agreement on the number of species in this Palearctic group has been reached, and bacula of only three or four of the numerous Old World subspecies have been figured. I have examined none.

Specimens examined: Two, from Wyoming; 42454 (31 mi. N Pinedale, 8025 ft., Sublette Co.), 37903 (23-1/2 mi. S, 5 mi. W Lander, 8600 ft., Fremont Co.).

***Microtus (Chilotus) oregoni* (Bachman)**

Fig. 45

Baculum: Stalk broad, greatest length (2.2 mm.) $1\frac{3}{4}$ times greatest breadth, $3\frac{1}{2}$ times greatest depth; three well-developed ossified processes; median process $\frac{2}{5}$ length of stalk, rounded or tapered terminally, proximal end opposed to tip of stalk and flattened obliquely; lateral processes $\frac{2}{3}$ length of median process, deeper than wide, curved; tuberosities of stalk well developed, confluent medially, visible in dorsal view; in end-view dorsal concavity narrow, moderately deep, rounded, ventral concavity wide, deep, flattened; base wider ventrally than dorsally; shaft tapering more or less uniformly, terminally inflated.

In the relative sizes, to each other and to the stalk, of the three digitate ossifications *M. oregoni* resembles closely the Old World representative of the same subgenus, *M. (Chilotus) socialis*, as figured by Argyropulo (1933b:181). In *M. oregoni* the greatest width of the baculum is more proximal on the stalk than in the *M. socialis* figured by Argyropulo but closely resembles the baculum of the *M. socialis* figured by Didier (1954:242). In possessing a shallow emargination in the base of the stalk and in possessing a median process that is smaller than the lateral processes, *M. socialis*, as figured by Didier, differs from *M. oregoni*. The baculum figured by Argyropulo (*loc. cit.*) of *Sumeriomys colchicus schidlovskii* [= *Microtus (Chilotus) socialis schidlovskii* according to Ognev, 1950:392] differs from other *Chilotus* that have been studied in having an unusually

elongate median process and a more distal placement of the widest part of the stalk.

Specimens examined: Three, of the subspecies *M. oregoni oregoni*, from 5 mi. N Orick, Humboldt Co., California, 3-C-248, collection of W. B. Quay; from Mary's Peak, Benton Co., Oregon, 66, collection of F. W. Sturges; and from Sec. 3, T. 11S, R. 5W, Benton Co., Oregon, 79183.

Microtus (Stenocranius) gregalis (Pallas)

Fig. 34

Baculum: Length of stalk (2.4 mm.) $1\frac{3}{4}$, times greatest breadth, $4\frac{1}{3}$ times greatest depth; median ossified process well developed, more than $\frac{1}{3}$ length of stalk, higher than wide, slightly bowed, closely appressed to terminus of shaft; basal tuberosities of stalk moderately developed, confluent medially, posterior profile of medial apex rounded in dorsal view, lateral indentations present, hence trilobate outline; in proximal end-view base wider ventrally, ventral concavity broader than dorsal concavity but of equal depth, medial constriction $\frac{2}{3}$ greatest depth; shaft slender in distal part, inflated terminally, and wider than high at mid-point of stalk; lateral profile a smooth slope of gradually decreasing curvature from point of greatest width to near distal end.

The baculum of this species figured by Ognev (1950:461) differs in having lateral ossified processes, and a more rounded base of the stalk. Resemblance to the New World *Stenocranius* is discussed below.

Specimen examined: One from "Eastern Europe," 8059.

Microtus (Stenocranius) miurus Osgood

Figs. 32 and 33

Baculum: Length of stalk (2.8 mm.) $1\frac{1}{2}$ times greatest breadth, $3\frac{1}{2}$ times greatest depth; median process ossified, $\frac{2}{5}$ to $\frac{3}{5}$ length of stalk, laterally

compressed, sometimes arched in dorsoventral plane; lateral processes cartilaginous, slender; basal tuberosities well developed, averaging less enlarged than shown in Figure 32, but more angular in lateral outline than shown in Figure 33; tuberosities confluent posteriorly; posterior profile smoothly rounded to trilobate, curvature at point of greatest breadth usually acute; in proximal end-view base wider dorsally, deep dorsal concavity, shallow ventral concavity, medial constriction $\frac{3}{5}$ of greatest depth; shaft slender anteriorly, at mid-point of stalk twice as wide as high, at tip higher than wide, laterally inflated; lateral profile in most specimens abruptly curved anterior to point of greatest breadth.

The single specimen of the Old World *M. (Stenocranius) gregalis* examined resembles the New World *M. (Stenocranius) miurus* in the angular lateral profile at the point of greatest breadth of the stalk, slender shaft in comparison to broad base of stalk, and presence of a single well-developed laterally compressed median process. The base of the stalk in the baculum of *M. gregalis* is less well developed and smaller than in the baculum of *M. miurus*.

Specimens examined: Nine, all of the subspecies *Microtus miurus muriei*, from the Brooks Range, Alaska; 51077 (Lake Schrader, 145°09'50", 69°24'28", 2900 ft., Romanzof Mts.); 51151, 51152, 51154, 51164, 51166, 51169 (last 6 from Wahoo Lake, 69°08', 146°58', 2350 ft.); 51210, 51213 (last 2 from Porcupine Lake, 68°51'57", 146°29'50", 3140 ft.).

***Microtus (Chionomys) nivalis* Martins**

Fig. 47

Baculum: Greatest length of stalk (2.7 mm.) $2\frac{1}{4}$ times greatest breadth, $4\frac{1}{2}$ times greatest depth; three digitate processes, lateral processes mostly cartilaginous in single adult examined; median process well ossified, approximately $\frac{1}{3}$ length of stalk, basally notched, not arched, laterally compressed distally; base of stalk broad and flat, basal tuberosities well developed, separate; posterior profile in dorsal view rounded, convex except for medial notch separating tuberosities; dorsal and ventral concavities deep, broad, equal; medial constriction less than $\frac{1}{2}$ greatest

depth; in dorsal view shaft tapering gradually from widest point, terminally rounded; at mid-point of stalk almost twice as wide as high.

In the elongate, largely cartilaginous lateral processes of the baculum, the specimen described above resembles *M. longicaudus*. The size of the median process in comparison to the size of the stalk is also the same. The lateral processes have larger ossifications and the base of the stalk is more robust in *M. longicaudus* than in *M. nivalis*.

The well ossified lateral processes and enlarged base of Didier's (1954:240) specimen suggest that it is of a more mature individual than the one described above. These specimens of *M. nivalis*, as well as the specimens of *M. longicaudus*, exhibit dorso-ventral flattening of the mid-part of the base of the stalk.

The baculum of a specimen from Switzerland is weakly developed, of small size (shaft 2.0 mm. in length), slender, thin, spatulate, and terminally inflated. Digital processes were not observed, perhaps owing to excessive maceration in preparation. The general appearance of the baculum is that of an immature individual, although the animal was not small (165 mm. total length in preservative).

Specimens examined: Two *Microtus nivalis nivalis*; Zermatt, Valais, Switzerland, 67105; Wetterstein, Germany, 65127.

***Microtus (Chionomys) longicaudus* (Merriam)**

Fig. 48

Baculum: Base of stalk well developed, greatest length (3 mm.) $1\frac{3}{4}$ times greatest breadth, $\frac{3}{2}$ times greatest depth; three ossified processes; base of median process rounded; median process slightly curved in dorsoventral plane, in length almost $\frac{1}{3}$ greatest length of stalk; ossifications in lateral processes variable in size, frequently widely separated from shaft by cartilage, rarely as large as median ossification; basal tuberosities usually well-developed, medially confluent; profile of base in dorsal view trilobate or irregularly convex throughout; constriction $\frac{1}{2}$ greatest depth; shaft

relatively straight or slightly bowed ventrally or dorsally, shaft at mid-point of stalk wider than high; tip of shaft laterally inflated; widest point of stalk approximately $\frac{1}{4}$ length of stalk from proximal end; lateral profile in dorsal view tapers gradually onto shaft anteriorly from point of greatest width of stalk; shaft variable, from slender terminally and nearly parallel sided (Fig. 48), to broad distally and tapered.

In many of the features that distinguish *M. longicaudus* (and the closely related insular species *M. coronarius*) from other North American *Microtus*, *longicaudus* resembles the Old World species of the subgenus *Chionomys* (that is to say, *M. nivalis*, *M. gud*, and *M. roberti*). These features are medium size, long tail, grayish color, montane habitat, relatively short molar tooth-row, moderate sized and unconstricted incisive foramen, relatively decurved upper incisors, elongate nasals, relatively broad interorbital region without well-developed median ridge, and similar chromosomes (Matthey, 1955:178). For these reasons I am here referring *Microtus longicaudus* to the subgenus *Chionomys*; previously it has not been referred to that subgenus.

Specimens examined: Six, of three subspecies; *Microtus longicaudus littoralis*, Sullivan Island, Alaska, 42972, 42969; *M. l. mordax*, 3/4 mi. N, 2 mi. W Allenspark, 8400 ft., Boulder Co., Colorado, 50335, 76829; *M. l. sierrae*, Crane Flat, Mariposa Co., California, 50252, 50253.

***Microtus arvalis* (Pallas)**

Fig. 22

Baculum: In the single specimen examined, stalk small, greatest length (2.3 mm.) $2\frac{1}{3}$ times greatest width, almost 6 times greatest depth, flattened proximally; three well-developed digitate processes, the median one ossified, the lateral processes cartilaginous; median ossification laterally compressed and decurved at tip, bilobate at base; basal tuberosities of stalk weakly developed, medially confluent; posterior profile in dorsal view evenly rounded; ventral concavity deeper and narrower than dorsal concavity, but both comparatively shallow; medial constriction $\frac{2}{3}$ greatest depth; shaft straight, at mid-point twice as wide as deep; lateral profile

tapering from greatest width gradually to parallel sides of distal third of stalk.

From the baculum of *Microtus arvalis* figured by Ognev (1950:173), and from the baculum figured by Didier (1954:238) my specimen differs in the absence of lateral ossifications in the digitate processes, smaller and slenderer median ossification, and weaker base. These differences in part may be owing to a difference in age, my specimen being the less mature.

Specimen examined: One from Vidy, Switzerland, 67101.

***Microtus orcadensis* Millais**

Fig. 24

Baculum: In the one specimen examined, stalk broad, greatest length (2.6 mm.) $1\frac{1}{2}$ times greatest breadth, $3\frac{1}{2}$ times greatest depth; three digitate processes ossified; median process relatively broad, in length more than $\frac{1}{2}$ length of stalk, triangular in dorsal view, with small spurs posterolaterally, middorsal ridge posteriorly; lateral ossifications slightly curved, slenderer, less than $\frac{1}{2}$ depth and less than $\frac{1}{2}$ transverse thickness of median process; basal tuberosities well-developed, confluent medially; in end-view base wider dorsally than ventrally, dorsal concavity broader and more abruptly curved at mid-point than ventral concavity; constriction $\frac{1}{2}$ greatest depth; posterior profile in dorsal view notched, setting off a posterior shelf; stalk including shaft wider than deep throughout, at mid-point width twice depth; lateral profile abruptly curved anterior to point of greatest width, sides of shaft tapering gradually anteriorly to rounded uninflated tip.

The baculum of this insular species, placed in the "*arvalis*" group by Ellerman (1941:595), resembles the baculum of both *Microtus agrestis* and *Microtus guentheri* more than it resembles the baculum of *Microtus arvalis*. Similarities in the chromosomes of *M. arvalis* and *M. orcadensis* were noted by Matthey (1953:254, 279), who was of the opinion that *M. orcadensis* is an insular derivative of the *arvalis*-group.

Specimen examined: One from the Orkney Islands, 67106.

Microtus guentheri Danford and Alston

Fig. 23

Baculum: In the one specimen examined, stalk broad, greatest length (2.9 mm.) $1\frac{1}{2}$ times greatest breadth, $3\frac{1}{2}$ times greatest depth; three digitate processes ossified; median process slightly less than $\frac{1}{2}$ length of stalk, broad, dorsally curved; curved lateral ossifications shorter and more slender than median ossification; basal tuberosities well developed, angular, confluent across posterior border of projecting shelf; in end-view tuberosities projecting ventrolaterally from central shelf; dorsal surface at medial constriction flat, ventral surface broadly and deeply concave; posterior profile in dorsal view trilobate, central lobe formed by posteriorly flattened shelf, surface of attachment visible only on lateral lobes; at mid-point stalk almost twice as wide as deep, depth of shaft greater than width proximal to inflated terminus.

Specimen examined: One from Palestine, 67104.

Microtus fortis Büchner

Fig. 25

Baculum: Stalk large, greatest length (3.8 mm.) $1\frac{4}{5}$ times greatest breadth, $4\frac{1}{2}$ times greatest depth; three digitate processes ossified; median ossification almost $\frac{1}{3}$ length of stalk; lateral ossifications slender, smaller than median ossification; posterior profile of stalk in dorsal view trilobate, basal tuberosities well developed, confluent medially; in end-view dorsal concavity broader and deeper than ventral concavity; medial constriction pronounced (less than $\frac{1}{2}$ greatest depth); lateral profile at widest point of stalk convex, becoming abruptly concave as the flange of the basal tuberosities grades into the shaft, then gradually converging to narrowest point $\frac{1}{3}$ of length of stalk from the terminus; stalk wider than deep in proximal $\frac{2}{3}$, circular in cross section in terminal $\frac{1}{3}$, slight terminal inflation.

A specimen figured by Ognev (1950:297) has the same general proportions, slender lateral processes, and proximal placement of the point of greatest breadth.

Specimens examined: Two from Chipo-ri, Korea, 60443, 63841.

Microtus montanus (Peale)

Figs. 19, 20 and 21

Baculum: Stalk broad, greatest length (varying with subspecies from 2.3 to 3.1 mm.) $1\frac{1}{2}$ to $1\frac{3}{4}$ times greatest breadth, $3\frac{1}{3}$ to $4\frac{1}{3}$ times greatest depth; three ossified processes, median one largest, more than twice as wide and as deep as shorter, slenderer, lateral processes; median process laterally compressed distally except in one specimen in which moderately inflated distally, proximally enlarged in some specimens (Fig. 21) and $\frac{1}{3}$ to $\frac{2}{5}$ length of stalk; base broad, posterior profile in dorsal view evenly convex throughout, at widest point of stalk abruptly incurved; basal tuberosities moderately to strongly developed, medially confluent; in end-view base wider ventrally than dorsally, dorsal concavity slightly to much deeper than the nearly flattened ventral concavity; medial constriction $\frac{2}{3}$ to $\frac{4}{5}$ of greatest depth; shaft relatively slender, at mid-point of stalk slightly wider than high and $\frac{1}{4}$ as wide as base of stalk, terminally rounded or slightly inflated; lateral profile in dorsal view a gradual curve from point of greatest width anteriorly onto shaft.

The different subspecies figured show the essential characteristics of the species, differing primarily in size.

Specimens examined: Fourteen, of three subspecies; *Microtus montanus amosus*, $\frac{1}{2}$ mi. E Soldier Summit, Wasatch Co., Utah, 62241; *M. montanus fusus*, La Manga Pass, Conejos Co., Colorado, 42164; 5 mi. N, 26 mi. W Saguache, 9500 ft., Saguache Co., Colorado, 42307, 42315; 5 mi. N, 27 mi. W Saguache, 9350 ft., Saguache Co., Colorado, 42308; 5 mi. N, 28 mi. W Saguache, 9325 ft., Saguache Co., Colorado, 42309; 5 mi. S, 24 mi. W Antonito, 9600 ft., Conejos Co., Colorado, 42327, 42330; Prater Canyon, Mesa Verde National Park, Montezuma Co., Colorado, 69456, 69457,

69463; *Microtus montanus nanus*, 2 mi. N, 2 mi. W Pocatello, Bannock Co., Idaho, 57470, 57472; ¼ mi. N, 2 mi. W Allenspark, 8400 ft., Boulder Co., Colorado, 50330.

***Microtus townsendii* (Bachman)**

Fig. 41

Baculum: Stalk broad, greatest length (3.0 mm.) $1\frac{1}{2}$ times greatest breadth, $4\frac{1}{2}$ times greatest depth; three ossified processes, median one largest, deeper and more than twice as wide as curved, shorter, compressed lateral processes and more than $\frac{2}{5}$ as long as stalk; base broad, in dorsal view posterior profile trilobate, basal tuberosities visible; basal tuberosities well developed, medially confluent; in end-view base wider ventrally than dorsally, dorsal concavity deeper than ventral concavity; medial constriction $\frac{3}{5}$ of greatest depth; shaft broad, at mid-point more than twice as wide as high and $\frac{1}{3}$ as wide as base of stalk, terminally rounded.

Specimens examined: Three, all *M. t. townsendii*; Fort Lewis, Pierce Co., Washington, 57998, subadult; Sec. 33, T. 11S, R. 5W, Benton Co., Oregon, 79186; Sec. 5, T. 12S, R. 4W, Benton Co., Oregon, 79188.

***Microtus oeconomus* (Pallas)**

Fig. 44

Baculum: Stalk broad and flattened, greatest length (3.5 mm.) $\frac{12}{3}$ to 2 times greatest width, 4 to $5\frac{1}{2}$ times greatest depth; three ossified processes, median one largest, lateral processes slender, relatively small; length of median process $\frac{3}{8}$ length of stalk; median process decurved, dorsoventrally flattened in some specimens, widened at base; attachment of processes to shaft displaced ventrally; base of stalk widened, posterior profile in dorsal view usually trilobate, in a few cases rounded, median lobe forming posterior shelf, lateral lobes dorsally raised and forming margins of lateral tuberosities; in end-view thickness frequently more or less uniform

throughout central part, broad depression dorsally, ventral concavity narrower and shallower (as figured); base, and occasionally shaft, flattened, width at mid-point of stalk 2 to 3 times depth, narrowest point posterior to terminal inflation of shaft in terminal $\frac{1}{3}$ of shaft.

The baculum of *M. oeconomus* (Old World) figured by Ognev (1950:257) resembles but exceeds that of *M. oeconomus* (New World) in the relatively large median process and slender lateral processes, but differs noticeably in the presence of a deep median notch in the base of the stalk. A specimen from Hungary is intermediate between Ognev's specimen and those from the New World in both size of median process and size of lateral processes, and has an unnotched base resembling that in Figure 44.

Specimens examined: Ten, of three subspecies; *M. oeconomus gilmorei*, Umiat, Alaska, 51354, 51361, 51399, 51408; Lake Schrader, Brooks Range, Alaska, 51422; *M. o. macfarlani*, 5 mi. NNE Gulkana, Alaska, 43039, 43041; 20 mi. NE Anchorage, Alaska, 43044; Kelsall Lake, British Columbia, 43048; *M. o. mehelyi*, Kisbalatan, Hungary, 75159.

***Microtus mexicanus* (Saussure)**

Figs. 35 and 36

Baculum: Stalk attenuate, greatest breadth relatively near proximal end; greatest length (3.1 to 3.4 mm.) more or less twice greatest breadth, 4 to 5 times greatest depth; usually a single process ossified; lateral processes relatively small, cartilaginous or (in three specimens, 63094, 69453, 68019) with small ossifications; median process relatively small, sometimes appressed to tip of shaft, in length less than $\frac{1}{4}$ length of stalk; posterior profile in dorsal view rounded, flattened posteriorly, or in some specimens trilobate with angular edges; in end-view relative depths of dorsal and ventral concavities variable, dorsal usually deeper than ventral; distal end of stalk frequently bowed dorsally; shaft slender distally, sometimes slightly inflated terminally, or (in one specimen, 63085) near tip small lateral projections that are perhaps fused lateral ossifications; lateral profile in dorsal view a gradual slope anteriorly from point of greatest width to slender tip.

Specimens examined: Thirteen, of four subspecies; *Microtus mexicanus mexicanus*, Las Vigas, Veracruz, 30692; Nevada de Toluca, México, 63101; Valle de Bravo, México, 63094; *Microtus mexicanus mogollonensis*, Mt. Taylor, Valencia Co., New Mexico, 63298, 76830; Park Well, Mesa Verde National Park, Montezuma Co., Colorado, 69448, 69453; Upper Nutria, McKinley Co., New Mexico, 69997, 70000; *Microtus mexicanus phaeus*, Sierra Patamba, 9000 ft., Michoacán, 63085; *Microtus mexicanus subsimus*, 2 mi. E Mesa de Tablas, Coahuila, 58916; 13 mi. E San Antonio de las Alazanas, Coahuila, 68019, 68021.

***Microtus californicus* (Peale)**

Fig. 37

Baculum: Stalk elongate, greatest length (3.0 mm.) $2\frac{1}{3}$ times greatest breadth, $4\frac{1}{2}$ times greatest depth; median process ossified, $\frac{1}{4}$ length of stalk, basally broadened, flattened and shallowly grooved ventrally to fit tip of shaft, to which the process is closely appressed; lateral processes cartilaginous; ends of stalk bowed upwardly; posterior profile of base of stalk rounded or slightly trilobate if posterolateral concavities form in tuberosities; moderate development of tuberosities, in end-view dorsal concavity slightly deeper and narrower than ventral concavity, both comparatively shallow, median constriction $\frac{4}{5}$ greatest depth; shaft curved, more or less terete at mid-point of stalk, terminally inflated dorsally; lateral profile in dorsal view gradually curved from point of greatest width anteriorly onto shaft.

Specimens examined: Two, of two subspecies; *Microtus californicus californicus*, 1 mi. NE Berkeley, in Contra Costa Co., California, 76828; *Microtus californicus mohavensis*, $\frac{1}{2}$ mi. SE Victorville, San Bernardino Co., California, 63745.

***Microtus pennsylvanicus* (Ord)**

Figs. 14, 15, 16 and 17

Baculum: Stalk heavy, broad, greatest length (2.2 to 3.0 mm.) $1\frac{1}{3}$ to $1\frac{2}{3}$ times greatest breadth, up to $3\frac{3}{4}$ times greatest depth; three ossified processes, median one largest, usually not twice so deep as lateral ossifications; median process usually distinctly widened basally, in length approximately $\frac{1}{2}$ length of stalk; base broad, frequently angular laterally and basally, sometimes bilobate; basal tuberosities well developed, medially confluent; in end-view more or less uniformly biconvex or ventral surface more flattened than dorsal surface, medial constriction $\frac{1}{2}$ to $\frac{2}{3}$ greatest depth; shaft relatively heavy, at mid-point stalk almost twice as wide as deep and $\frac{1}{3}$ as wide as base of stalk; shaft terminally rounded and sometimes slightly inflated; lateral profile in dorsal view abruptly or gradually curved anterior to point of greatest width and then gradually curved anteriorly.

Specimens examined averaged slightly smaller and were more variable than those described by Hamilton (1946:382). The greater variation may be in part geographic, as five subspecies are represented. Lateral processes are the last to ossify. One specimen (75082) with well-ossified median process lacks any lateral ossification. Four bacula of *M. pennsylvanicus* (locality not specified) studied by Dearden (1958:547) agree in general with the description above.

One specimen shows a break, perhaps resulting from injury, in the shaft (Fig. 14). One specimen has a posteromedian spine on the median digital ossification (Fig. 16). Comparison with *M. agrestis* is included with the description of *M. agrestis*.

Specimens examined: Thirteen, of six subspecies; *Microtus pennsylvanicus alcorni*, 20 mi. NE Anchorage, Alaska, 43043; *Microtus pennsylvanicus finitus*, Laird, Yuma Co., Colorado, 68544; *Microtus pennsylvanicus modestus*, 5 mi. N, 26 mi. W Saguache, 9500 ft., Saguache Co., Colorado, 42306; 3 mi. N, 16 mi. W Saguache, 8500 ft., Saguache Co., Colorado, 42416, 42417, 42418; 1 mi. S, 2 mi. E Eagle Nest, 8100 ft., Colfax Co., New Mexico, 42430, 42439; *Microtus pennsylvanicus pennsylvanicus*, 2 mi. S, 3 mi. E Ft. Thompson, 1370 ft., Buffalo Co., South Dakota, 42379; Vermillion, Clay Co., South Dakota, 37070; *Microtus pennsylvanicus pullatus*, 12 mi. S, 5 mi. E Butte, Silver Bow Co., Montana, 57501, 57503;

Microtus pennsylvanicus uligocola, Muir Springs, 2 mi. N, 2½mi. W Ft. Morgan, Morgan Co., Colorado, 75082.

Microtus agrestis (Linnaeus)

Fig. 18

Baculum: Greatest length of stalk (2.9 mm.) twice greatest breadth, 4½ times greatest depth; stalk well developed, shaft not flattened dorsoventrally; large median ossified process, minute lateral ossifications in single specimen examined; length of stalk 2½ times length of median ossification which is higher than wide, slightly decurved, sagittate in dorsal view, with three-cornered base; basal tuberosities of stalk moderately well developed, medially joined; posterior profile in dorsal view evenly rounded; ventral concavity broader than, but of comparable depth to, dorsal concavity in end-view, base of stalk wider ventrally, constriction $\frac{3}{4}$ greatest depth; at mid-point of stalk shaft is but slightly wider than high; pronounced terminal inflation of shaft; lateral profile in dorsal view sloping abruptly from widest point of stalk anteriorly onto stalk which then tapers more gradually to terminal inflation.

From the baculum of its New World counterpart, namely *Microtus pennsylvanicus*, my specimen of *Microtus agrestis* and the specimen figured by Didier (1954:239) differ in their minute lateral processes, relatively larger median processes, and more elongate, less dorsoventrally flattened shafts.

The specimen of *M. agrestis* figured by Ognev (1950:320), in dorsal view has lateral concavities producing a somewhat trilobate outline in the base of the stalk, and the lateral processes are well developed; the median process is larger and bulbous, wider distally than proximally. Without larger numbers of bacula of *M. agrestis* I am unable to reconcile these differences. The differences between *M. agrestis* and *M. pennsylvanicus* seem comparable to the differences between some other species of *Microtus*.

Specimen examined: One, from Gryon, Switzerland, 67102.

Microtus (Pedomys) ochrogaster (Wagner)

Fig. 31

Baculum: Stalk broad, greatest length (3.2-4.0 mm.) $1\frac{2}{3}$ to 2 times greatest breadth, $2\frac{1}{2}$ to 4 times greatest depth; median process ossified, relatively small, less than $\frac{3}{10}$ length of stalk; lateral processes arising from subterminal part of stalk, cartilaginous or with small ossifications; posterior profile in dorsal view broadly rounded or slightly angular, widest point of stalk $\frac{1}{6}$ to $\frac{1}{4}$ the length of stalk from base; basal tuberosities well developed and medially confluent, in end-view dorsally convex, or at least less deeply concave than ventrally; shaft straight, base bent ventrally or more commonly dorsally; at mid-point of stalk wider than high, often twice as wide as high; viewed from above, lateral profile from point of greatest breadth to middle of shaft a gradual sigmoid curve; slight terminal inflation of shaft.

Specimens examined: Forty-one, of three subspecies; *Microtus ochrogaster haydeni*, Muir Springs, 2 mi. N, $2\frac{1}{2}$ mi. W Ft. Morgan, Morgan Co., Colorado, 74995, 74998, 74999, 75002; 1 mi. W Laird, Yuma Co., Colorado, 57304, 76833; 2 mi. N, 2 mi. W Haigler, Dundy Co., Nebraska, 75016; 2 mi. S Franklin, Franklin Co., Nebraska, 75043, 75044; Atwood, Rawlins Co., Kansas, 75020, 75023, 75025, 75027, 75028; 1 mi. N, 2 mi. E Oberlin, Decatur Co., Kansas, 75030, 75032, 75034, 75035, 75036; $1\frac{1}{2}$ mi. N, $\frac{1}{4}$ mi. E Norton, Norton Co., Kansas, 68327; 1 mi. SW Norton, Norton Co., Kansas, 75037; 2 mi. S, 1 mi. W Norton, Norton Co., Kansas, 75038; *M. ochrogaster ochrogaster*, Rydal, Republic Co., Kansas, 75047-75053, 75060, 75062, 75063, 75066, 75070, 75071, 75073; 1 mi. N, 1 mi. W Holton, Jackson Co., Kansas, 75077; 2 mi. W Court House, Lawrence, Douglas Co., Kansas, 76832; Univ. Kansas Natural History Reservation, Douglas Co., Kansas, 68536; *M. ochrogaster taylori*, Meade County State Park, Kansas, 68539, 68542.

Microtus (Pitymys) pinetorum (LeConte)

Figs. 27 and 28

Baculum: Stalk broad, greatest length (2.5 to 2.7 mm.) $1\frac{2}{3}$ times greatest breadth, 4 times greatest depth; median process ossified, size small, $\frac{1}{5}$ length of stalk, higher than wide, having small anterodorsal prominence in both specimens examined; lateral processes cartilaginous, relatively small, displaced posteriorly, attenuate; posterior margin in dorsal view broadly rounded, or having blunt median apex, convex throughout; basal tuberosities moderately well developed, medially confluent, barely visible in dorsal view when mature; in end-view median constriction $\frac{4}{5}$ greatest depth, ventral concavity deeper than dorsal concavity, both comparatively shallow; stalk at mid-point $1\frac{1}{2}$ times as wide as deep; shaft relatively slender, bowed dorsally at tip, relatively straight otherwise; lateral profile in dorsal view a gradual concave slope from point of greatest width anteriorly to distal part of shaft.

Specimens examined: Two, from Douglas Co., Kansas, 76834 (2 mi. N Baldwin), 68545 (1 mi. NE Pleasant Grove).

***Microtus (Pitymys) parvulus* (Howell)**

Fig. 40

Baculum: Stalk broad, greatest length (2.4 mm. in specimen examined) $1\frac{3}{4}$ times greatest breadth, 4 times greatest depth; median process ossified, size small, less than $\frac{1}{4}$ length of stalk, wider than high, terminally flattened; lateral processes cartilaginous, relatively small, attenuate; posterior margin in dorsal view flattened, irregularly curved with concavities medially and laterally; basal tuberosities well developed, medially confluent; visible in dorsal view; in end-view median constriction $\frac{2}{3}$ greatest depth, ventral concavity well-formed, no dorsal concavity; stalk at mid-point twice as wide as deep; shaft relatively slender, bowed dorsally toward tip; in dorsal view lateral profile a gradual concave slope from point of greatest width anteriorly to distal part of shaft; tip of shaft enlarged.

The baculum of *M. parvulus* resembles that of *M. pinetorum* more than it resembles the baculum of any other microtine studied, differing primarily in smaller size.

Specimen examined: One, from 1 mi. W Micanopy, Alachua Co., Florida, Univ. Florida No. 1508.

***Microtus (Pitymys) quasiater* (Coues)**

Figs. 29 and 30

Baculum: Stalk broad, greatest length (2.6-3.2 mm.) $1\frac{1}{3}$ to $1\frac{2}{3}$ times greatest breadth, $3\frac{1}{3}$ to $3\frac{2}{3}$ times greatest depth; median process ossified, with ventral depression, process $\frac{1}{4}$ to $\frac{1}{3}$ length of stalk, appressed to tip of shaft, wider than high proximally, relatively broad terminally; lateral processes cartilaginous, small, attenuate; posterior profile of stalk in dorsal view broadly rounded, bilobate, or trilobate, median lobe formed by posterior projection of dorsal shelf between enlarged lateral tuberosities that form outer lobes, posterolateral faces of these tuberosities visible in dorsal view of stalk; in end-view dorsal surface slightly concave, ventral concavity broad and deep, median constriction $\frac{1}{2}$ greatest depth; shaft flattened except tip that is more terete, and bowed dorsally; at mid-point, stalk twice as wide as high; shaft relatively slender terminally, narrower than median ossification.

The baculum of *M. quasiater* is the largest and has the best developed base and median process of the three American species of the subgenus *Pitymys*. The three species closely resemble each other in basic form.

Specimens examined: Five, all from Veracruz; Teocelo, 4500 ft., 30709, 30711; 4 km. N Tlapacoyán, 1700 ft., 24466; 5 km. N Jalapa, 4500 ft., 19869, 19878.

***Microtus (Pitymys) fatioi* (Mottaz)**

Fig. 26

The baculum of a single specimen (KU 67103) of *M. fatioi* from Zermatt, Valais, Switzerland, was examined. The baculum is immature, as evidenced

by its small size, slender stalk and absence of ossified processes, therefore no characterization is included.

The baculum of another Old World species of the subgenus *Pitymys*, *M. pyrenaicus* from France, figured and described by Didier (1954:242-243), differs from all New World *Pitymys* examined in possessing ossified lateral processes.

The status of *Pitymys*, as a genus or as a subgenus, is uncertain. Hall and Cockrum (1953:448) considered the North American *Pitymys* and *Pedomys* as subgenera of *Microtus*. They did not state specifically the basis for this point of view, but mention the fact that these two subgenera (*Pitymys* and *Pedomys*) closely resemble each other cranially. These authors did not study nor comment upon the status of the Old World *Pitymys*. It may be asked whether the Old World and New World *Pitymys* have developed as fossorial *Microtus* independently, or from an ancestor common to both groups and not common to any other *Microtus*. Matthey (1955:202) found 62 chromosomes (2N) in both the New World *Pitymys pinetorum* and the Old World *Pitymys duodecimcostatus*. This suggests, but does not prove, common ancestry.

Neofiber alleni True

Fig. 49

Baculum: Stalk massive, greatest length (4.7 mm.) $1\frac{3}{4}$ times greatest breadth, 4 times greatest depth; ossification in digitate processes variable; in one (KU 27123) of two specimens examined lateral processes ossified and median process unossified, as in two specimens examined by Hamilton (1946:379) from "southern Florida"; in my other specimen (KU 27268) that is possibly more mature, median process ossified although less deeply stained than lateral ossifications or stalk; posterior profile in probable dorsal view roughly rounded; in end-view probable dorsal concavity deep, ventral concavity broad but shallow, and with center convex; median constriction $\frac{3}{5}$ greatest depth; shaft heavy, least depth $\frac{2}{3}$ greatest depth of base; stalk, at mid-point, slightly wider than deep and more than $\frac{1}{3}$ width of base; lateral profile in dorsal view sharply incurved distal to point of greatest

breadth, shaft therefore relatively distinct from basal part of stalk; slight subterminal constriction; tip less reduced in the two specimens examined than in two figured by Hamilton. In preparation, the tissues that make it possible to distinguish with certainty the dorsal and ventral surfaces of the baculum were removed in both specimens.

Specimens examined: Two, of the subspecies *Neofiber alleni alleni*, 2 mi. S Gainesville, Alachua Co., Florida, 27268; 1 mi. E Courtenay, Merritt Island, Brevard Co., Florida, 27123.

Lagurus curtatus (Cope)

Fig. 46

Baculum: Stalk slender, greatest length (2.5 mm.) 2 to $2\frac{2}{3}$ times greatest breadth, 4 to 5 times greatest depth; three ossified processes; median one more than $\frac{1}{3}$ length of stalk, curved dorsally toward tip, proximally flattened and having acute lateral angles in dorsal view, wider than deep except in distal half; lateral processes smaller than median one, slenderer, shorter, of approximately same depth, also curved dorsally; base of stalk well developed, basal tuberosities medially confluent, in part visible in dorsal view, in end-view wider ventrally than dorsally, dorsal and ventral concavities of equal depth and both wide, medial constriction $\frac{1}{2}$ greatest depth; posterior profile in dorsal view broadly bilobate; lateral profile with abrupt transition from basal tuberosities to gradually converging, slightly curved sides of shaft; shaft terminally inflated.

Dearden (1958:543) described and figured the bacula of six subspecies of *Lagurus curtatus* and two Asiatic species, *Lagurus lagurus* and *Lagurus luteus*. He examined at least 34 specimens of *L. curtatus* and found geographic variation in size, breadth of shaft distally, and proportions of digital ossifications to each other and to the stalk. The description that I have given above pertains to *L. c. levidensis*.

The baculum of the Asiatic *Lagurus (Lagurus) lagurus* figured by Ognev (1950:554) agrees with that of *Lagurus (Lemmiscus) curtatus*, described here, in the relatively elongate shaft and slender stalk, the proportions of the

processes, and the well-formed and moderately enlarged base of the stalk. The bacula of three *Lagurus lagurus* examined by Dearden (1958:545) were of older individuals than the specimen that Ognev figures and differ from it and from bacula of *Lagurus curtatus* (all subspecies) in the unusual, almost heart shaped, median process, and in larger size. *Lagurus luteus* examined by Dearden (1958:545) differs from both *Lagurus lagurus* and *Lagurus curtatus* in lacking lateral digital ossifications and in having shorter median digital ossifications and wider base of stalk.

Specimens examined: Seven *Lagurus curtatus levidensis* from Wyoming; 9 mi. S Robertson, Uinta Co., 26045, 26053; 8 mi. S, 2½ mi. E Robertson, Uinta Co., 26049; Farson, Sweetwater Co., 37906; 16 mi. S, 11 mi. W Waltman, Natrona Co., 42457; 32 mi. S, 22 mi. E Rock Springs, 42465, 42466.

The following key to the bacula in some adult North American Microtinae is intended to help point out some of the most important differences. It should be noted that not all species can be keyed out on the basis of the baculum. The most difficult group in this respect includes the species of *Microtus* that have small or no ossified lateral processes, especially species of the subgenera *Pedomys* and *Pitymys*, and the two species *Microtus californicus* and *Microtus mexicanus* of the subgenus *Microtus*. Another complicating factor is the variability of bacula evident in some species even in the small samples available. It is to be expected that additional specimens will show variations not yet observed.

KEY TO THE BACULA OF SOME NORTH AMERICAN MICROTINES

- | | |
|---|------------------------|
| 1. Length of lateral digital ossifications more than $\frac{1}{3}$ breadth of stalk | 2 |
| 1'. Length of lateral digital ossifications less than $\frac{1}{3}$ breadth of stalk or absent | 15 |
| 2. Size small (total length of baculum less than 5.5 mm.) | 3 |
| 2'. Size large (total length of baculum more than 5.5 mm.) | 14 |
| 3. Width at mid-point of stalk more than $\frac{1}{3}$ greatest breadth of stalk | 4 |
| 2'. Size large (total length of baculum more than 5.5 mm.) | 14 |
| 3'. Width at mid-point of stalk less than $\frac{1}{3}$ greatest breadth of stalk, | 8 |
| 4. Stalk, viewed from proximal end hour-glass shaped, and width of stalk less than $\frac{1}{2}$ length of stalk.... <i>Phenacomys intermedius</i> , | p. 197 |
| 4'. Stalk not both hour-glass shaped when viewed from proximal end, and with width less than $\frac{1}{2}$ length of stalk | 5 |
| 5. Shaft thin basally, thickness less than $\frac{1}{3}$ of greatest breadth | 6 |
| 5'. Shaft thick basally, thickness $\frac{1}{3}$ or more of greatest breadth | 7 |
| 6. Stalk more or less straight, base not deflected. <i>Microtus oeconomus</i> , | p. 204 |
| 6'. Stalk spatulate, and base deflected from axis of shaft.... <i>Microtus guatemalensis</i> , | p. 198 |
| 7. Base enlarged, depth nearly $\frac{1}{2}$ of breadth.... <i>Lemmus trimucronatus</i> , | p. 193 |
| 7'. Base moderately enlarged, depth near $\frac{1}{3}$ of breadth.... <i>Microtus pennsylvanicus</i> , p. 206 , or <i>Microtus townsendii</i> , | p. 204 |

8. Base hour-glass shaped as viewed from proximal end.... <i>Phenacomys intermedius</i> ,	p. 197
8'. Not so	9
9. Lateral processes separated from tip of shaft by more than the thickness of the lateral process	10
9'. Lateral processes separated from tip of shaft by less than the thickness of the lateral process	11
10. Lateral processes more than ½ the width of median process.... <i>Microtus longicaudus</i> ,	p. 201
10'. Lateral processes slender, less than ½ the width of median process.... <i>Microtus montanus</i> ,	p. 204
11. Lateral ossifications equal in length to median ossification.... <i>Clethrionomys</i> ,	p. 194
11'. Lateral ossifications shorter than median ossification	12
12. Size small, less than 3.4 mm. in total length.... <i>Microtus oregoni</i> ,	p. 199
12'. Size medium, more than 3.4 mm. in total length	13
13. Greatest width of stalk at a point about 1/3 of length of stalk from base.... <i>Microtus chrotorrhinus</i> (Hamilton, 1946:382).	
13'. Greatest width of stalk at a point less than 1/3 of length of stalk from base.... <i>Lagurus curtatus</i> ,	p. 210
14. Size of baculum larger, base more than 3 mm. wide, processes all well developed.... <i>Ondatra zibethicus</i> ,	p. 198
14'. Size of baculum smaller, base less than 3 mm. wide, processes poorly developed in some animals.... <i>Neofiber alleni</i> ,	p. 209
15. At least one digital ossification present	16
15'. Digital ossifications not present.... <i>Dicrostonyx groenlandicus</i> ,	p. 193
16. Breadth of stalk at least ½ length of stalk	17
16'. Breadth of stalk less than ½ length of stalk	19
17. Length of stalk greater than 3.6 mm. and less than 1½ times its greatest breadth.... <i>Microtus richardsoni</i> ,	p. 199
17'. Length of stalk usually less than 3.6 mm., or if more than 3.6 mm. (up to 4.0 mm.) then length 1½ times or more its	18

greatest breadth

18. Median process attenuate distally in dorsal view, and relatively long (more than twice its own breadth), $\frac{1}{5}$ to $\frac{3}{5}$ the length of stalk; breadth of stalk usually $\frac{2}{3}$ or more length of stalk.... *Microtus miurus*, [p. 200](#)

18'. Median process relatively blunt distally in dorsal view, relatively short (usually less than $\frac{1}{4}$ length of stalk), breadth of stalk usually less than $\frac{2}{3}$ length of stalk.... *Pitymys*, [p. 208](#), *Pedomys*, [p. 207](#), or *Microtus mexicanus*, [p. 205](#)

19. Distal processes small and firmly ankylosed to distal end of shaft.... *Phenacomys longicaudus*, [p. 197](#)

19'. Distal processes if present not firmly ankylosed to distal end of shaft 20

20. Dorsal concavity of base as viewed from proximal end usually deeper than ventral concavity.... *Microtus mexicanus*, [p. 205](#)

20'. Dorsal and ventral concavities of base equal in depth or ventral one the deeper 21

21. Total length of baculum more than 3.6 mm.... *Microtus californicus*, [p. 205](#)

21'. Total length of baculum less than 3.6 mm.... *Synaptomys cooperi*, [p. 194](#)

DISCUSSION

Owing to shortness of lower incisors and present geographic distribution of the species, Hinton (1926:35) considered the Tribe Lemmi (lemmings) to be more primitive than the Tribe Microti (voles). The surviving lemmings are specialized in many features and therefore are considered as advanced end-products of an evolutionary radiation of a primitive microtine stock, of which all earlier stages are extinct.

Hinton regarded *Dicrostonyx* as the most primitive of the genera of lemmings on account of its more complex molar teeth (complexity was considered to be primitive), and on account of the presence of three primitive longitudinal rows of tubercles in unworn molars. The other three genera were arranged in order of increasing specialization as follows: *Synaptomys*, *Myopus*, *Lemmus*.

If the baculum tended to retain its primitive character while specializations in the external anatomy developed, and if the above arrangement is correct the most primitive bacula would be found in *Dicrostonyx* and in *Synaptomys*. The baculum in these two genera in comparison to that in *Myopus* (as figured by Ognev, 1948:512) and *Lemmus* has a slenderer stalk and smaller digital ossifications or none at all. The baculum in the genera of lemmings increases in robustness and the development of processes from *Dicrostonyx*, to *Synaptomys*, to *Myopus*, to *Lemmus*—the same order outlined above for total of specialization. The two extremes in this series are near the extremes of variation in bacula to be found in all microtines. The baculum in lemmings as a group cannot then be considered more primitive than in voles as a group, although the voles are usually considered to be more advanced. The situation in the voles, as we shall see, casts a different light on the matter.

The voles, Tribe Microti, were considered by Hinton (1926:40) to be more advanced than the lemmings because the incisors of the voles are longer

and the root of their last lower molar is lingual to the root of the incisor. Hinton thought also that the murine ancestors of microtines had shorter incisors and that the backward extension of the incisors in the voles is a more ancient feature than the hypsodonty of the molars. A trend in the molar teeth has been toward greater hypsodonty. The voles in which the molars are least hypsodont are thus considered primitive. These include the living genera *Clethrionomys*, *Phenacomys*, *Ondatra*, *Dolomys*, *Ellobius*, and *Prometheomys*. Therefore, the baculum, in these assumedly primitive genera, would be expected to resemble the baculum in the lemmings or at least the most primitive lemmings. This is not the case.

The bacula that I have examined of *Clethrionomys* and *Phenacomys* have well-developed digital ossifications. In this they resemble the baculum of the genus *Lemmus*, the most advanced genus of lemmings according to Hinton. The baculum of *Dolomys* has not been studied. The baculum in *Ondatra*, and in *Prometheomys* as illustrated by Ognev (1948:552), also possesses well-developed processes. The baculum of *Ellobius* is small and lacks processes (as figured by Ognev, 1950:662). No ossification was found in a single specimen of *Ellobius* examined by me although the entire glans penis was removed and cleared without dissection. So far as known then, with the exception of *Ellobius* and *Phenacomys longicaudus* (Dearden, 1958:547), the primitive microtines having rooted molars possess bacula having three well-developed ossified processes.

Voies of the genus *Microtus* vary in the structure of the baculum almost as much as do the lemmings. Within the single subgenus *Microtus* some individuals of *Microtus mexicanus*, for example, have minute ossified lateral processes and other individuals lack these processes; *Microtus pennsylvanicus* and some other species have proportionately large lateral ossifications. If the well-developed condition of the baculum in the microtines having rooted molars is primitive, then within the genus *Microtus* those species having well-developed bacula may be considered primitive.

The genera *Lagurus* and *Neofiber* have moderately developed or well-developed lateral processes. *Neofiber* exhibits a tendency, not prominent elsewhere, to have a proportionately smaller median process rather than reduced lateral processes.

American species of *Microtus* (genus and subgenus) that have moderately- to well-developed ossified lateral processes are *M. townsendii*, *M. oeconomus*, *M. pennsylvanicus*, *M. montanus*, and *M. chrotorrhinus*. *Microtus* of other subgenera having this type of baculum include *M. (Herpetomys) guatemalensis*, *M. (Chilotus) oregoni*, and *M. (Chionomys) longicaudus*.

American species of *Microtus* (genus and subgenus) in which the lateral ossifications are weakly developed or absent (although cartilaginous lateral processes are present) include *M. mexicanus* and *M. californicus*. In other subgenera, species of *Microtus* having reduced lateral ossifications are *M. (Pedomys) ochrogaster*, *M. (Pitymys) pinetorum*, *M. (Pitymys) parvulus*, *M. (Pitymys) quasiater*, *M. (Arvicola) richardsoni*, and *M. (Stenocranius) miurus*.

The microtines are essentially holarctic in distribution. Both of the tribes, the lemmings and the voles, as well as primitive representatives of each tribe (not considering *Ellobius*) occur in both the Old World and New World. It is not certain on which continent (or continents) the Microtinae first differentiated. It is certain, however, that at various times, both early and late in the evolution of the subfamily, representatives have crossed from Eurasia to North America or *vice versa*. Each of 10 or more microtines in the New World is more closely related to some microtine in the Old World than to any other microtine in the New World.

The similarities or differences in the baculum in Old World and New World representatives placed in the same genus or subgenus, or thought to be "companion species" have been commented upon in accounts of *Lemmus*, *Dicrostonyx*, *Clethrionomys*, *Lagurus*, *Arvicola*, *Stenocranius*, *Chilotus*, *Chionomys*, *Pitymys*, and in accounts of *Microtus agrestis* as compared with *M. pennsylvanicus*, and *Microtus oeconomus* (both Old World and New World).

The baculum in the Microtinae more closely resembles the baculum in the Cricetinae of the Old World than in the Murinae, or than in any other rodents known to me. This resemblance suggests relationship between Microtinae and Cricetinae.

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